CALIFORNIA STATE UNIVERSITY, NORTHRIDGE

Towards Physiological and Transcriptomic Analysis of Turmeric (*Curcuma longa* L.) Under Drought Stress

A thesis submitted in partial fulfillment of the

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Master of Science in Biology

By

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Dedication

For the endless love, support and encouragement, this thesis is dedicated to my family.

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Abstract

Towards Physiological and Transcriptomic Analysis of Turmeric (*Curcuma longa* L.) Under Drought Stress

By Bhiolina Bharadwaj

Master of Science in Biology

For nearly 4000 years, turmeric has been one of the most important medicinal plants (*Curcuma longa)*. In Southern Asia, it has been used as spices, food preservatives, beauty care products, and coloring material. Lately, unfavorable environmental stresses are hindering the growth and development of turmeric, leading to a great loss of its production in many parts of Southern Asia, including India. Out of all abiotic stresses, drought is one of the most concerning environmental stresses in areas around India. However, there are only a few studies about the effect of water scarcity as a stress on turmeric plants at a molecular level. In my research, I am investigating physiological and biochemical changes in turmeric plants under drought stress. Understanding differential expression of genes in turmeric plants under environmentally stressful conditions will be crucial to comprehending the entire mechanism responsible for drought stress tolerance in this medicinal plant. In this research study, physiological and molecular analyses were conducted after introducing artificial drought stress to turmeric plants by withdrawing watering for 21 days. Physiological studies involved photosynthetic rate, chlorophyll content, moisture content, and catalase activity estimation. While for molecular studies, RNA was extracted from drought-stressed and controlled plants, and then the Illumina Truseq kit was used to prepare barcoded cDNA libraries. Illumina Miseq sequencing system (Miseq v3) kit was used for sequencing the barcoded cDNAs from controlled and drought-stressed plants. It has been found that drought stress does affect turmeric plants

at the physiological level by reducing photosynthetic rates and chlorophyll content. One of the interesting findings is the downregulation of catalase activity in senescing plants under drought stress. Furthermore, with identified upregulated and downregulated genes from previous studies and physiological studies, we hypothesized the downregulation of the curcuminoid biosynthesis pathway and phenylpropanoid biosynthesis pathway, primarily due to the downregulation of auxin and the squamosa-promoter binding protein (SPL) family. Furthermore, this study provides a future opening to studying genetically improved turmeric plants with a tolerance to drought stress.

Chapter 1

Introduction

Since the early days, humans have involved plant-derived products in their daily life for various purposes (Prasad et al. 2011). From food to shelter, in every possible way, plant products have been used by ancient cultures. Some plant products have also been used in health care for their medicinal properties. These products are primarily secondary metabolites released by higher plants as a defense mechanism (Prasad et al. 2011). From that time, primitive cultures have identified plants with medicinal properties and used them to cure many ailments. Even now, Ayurvedic (an Ancient Indian medical system) practices plant-based medicines to treat various diseases, including cancers (Prasad et al. 2011). The World Health Organization has been trying to identify all plants with medicinal properties that exist worldwide (Farnsworth & Soejarto, 1988).

According to Ayurvedic, turmeric (*Curcuma longa)* from the Zingiberaceae family has a long history of therapeutic purposes. Besides its medicinal uses, turmeric has been used for flavor and aroma as well as to store food (Govindarajan et al. 1980). For decades, turmeric has been used in beauty products because of its bright color, which also gives the golden yellow color of Indian curry. In many Indian cultures, turmeric has also been used in religious ceremonies (Nair, 2019). Many ancient Indian physicians (such as Charaka and Sushruta) and Greek physicians (such as Dioscorides) mentioned turmeric for medicinal uses. The Egyptian and Middle East regions were aware of turmeric's pharmaceutical properties for thousands of years.

Morphology of turmeric

It is generally 3-5 feet tall and produces dull yellow flowers, but it is a sterile plant that does not produce any seeds. Hence it reproduces vegetatively. Plants above soil grow to full height with long petioles followed by large leaves (Figure 1). The underground stem is called a rhizome which is thick and fleshy. The rhizome is the part that is collected annually when it matures for further uses including medicinal uses, and new plants also sprout when these rhizomes are cultivated for the following growing season. It is also the part that can be dried and ground to make the powdered spice. The use of this spice is very versatile in India. A rhizome is tuberous and has dried and rough skin with segments on it. The color varies from dull yellow to brownish to pale orange (https://www.britannica.com/plant/turmeric). It sizes ranges from 1-3 inches in length and 0.5-1 inches in diameter. The ends of a mature rhizome appear to be pointed or tapered (Figure 2).

Figure 1: Turmeric (*Curcuma longa*: CL11 variety) from Alabama A&M University, grown in the greenhouses of Biology department, CSUN

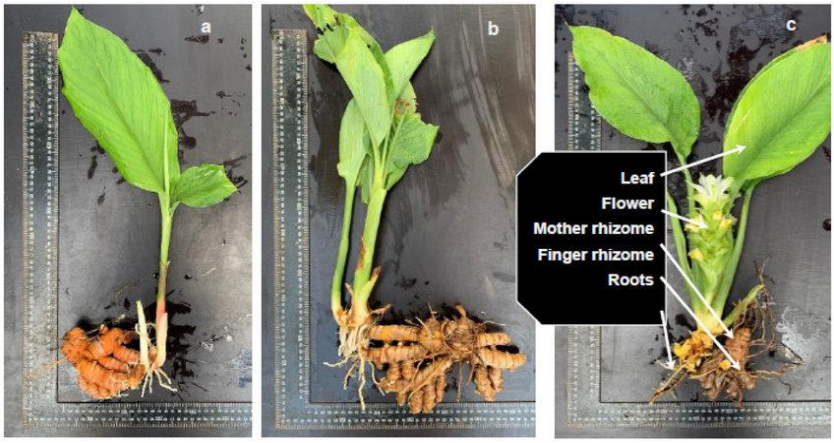


Figure 2: Morphology of turmeric plants (Source: Farrar et al. 2021)

Origin of turmeric

The origin of *Curcuma longa* is not conclusive yet. Recently, near New Delhi, India, pots from 2500 BCE uncovered traces of turmeric (https://www.pbs.org/food/the-history-kitchen/turmeric-history/). However, all the pieces of evidence that have been reported are not enough to prove that India is where turmeric first appeared (Nair, 2019).

Geographic Distribution of turmeric

Turmeric is primarily distributed in India and South-East Asia, and then it expanded into different regions (Figure 3). It reached China in the 7th century, East Africa in the 8th century, and West Africa in the 13th century (Pl@ntuse.org-turmeric). Then in the fourteenth century, European explorers in India introduced turmeric to the Western world (Nair, 2019).

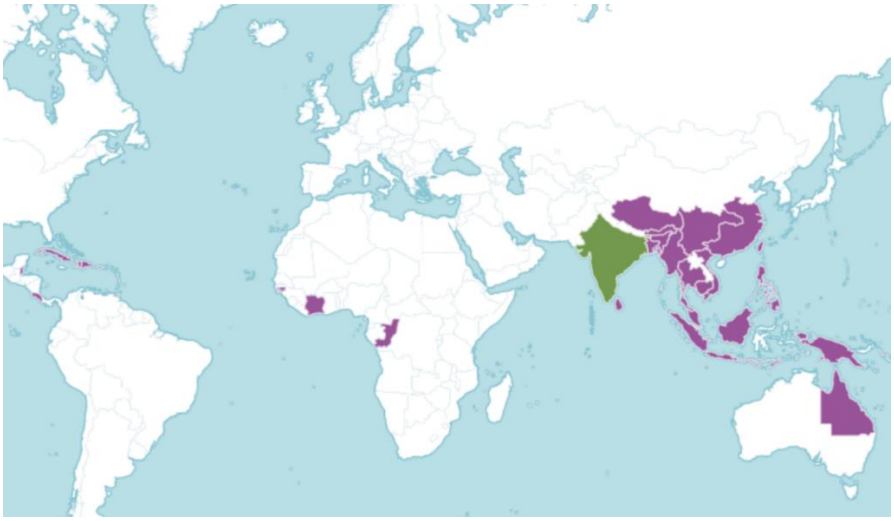


Figure 3: Turmeric distribution around the world (Source: powo.science.kew.org)

The different names of turmeric

The name turmeric derives from the Latin word terra merita, which means the meritorious earth (referring to the color of the ground turmeric) (Prasad et al. 2011). The international common name: Curcuma. Turmeric has several names depending on different cultures and languages (Table 1). In Sanskrit, turmeric has at least 53 different names, such as haridra, gauri, kanchani, haldi, tami, and aushadhi.

Table 1: Different Names of Turmeric in Different Languages

Language	Name
Arabic	Kurkum
Armenian	Toormerik
Assamese	Halodhi
Bengali	Halud
Chinese	Yu chin
English	Indian saffron
French	Curcuma

German	Curcuma
Hindi	Haldi
Japanese	Ukon

(Source: Prasad & Aggarwal,2011)

Phytochemical content in dried turmeric rhizomes

Turmeric contains many biochemical products which have a high demand in the industrial sector in western countries (Nair, 2019). It contains both volatile as well as non-volatile compounds. Many types of secondary metabolites such as terpenoids, curcuminoids, phenolic compounds, flavonoids, saccharides, steroids, fatty acids, and alkaloids, have already been investigated in essential oils obtained from steam distillation of turmeric rhizomes (Meng et al. 2018 & Nisar et al. 2015) (Figure 4).

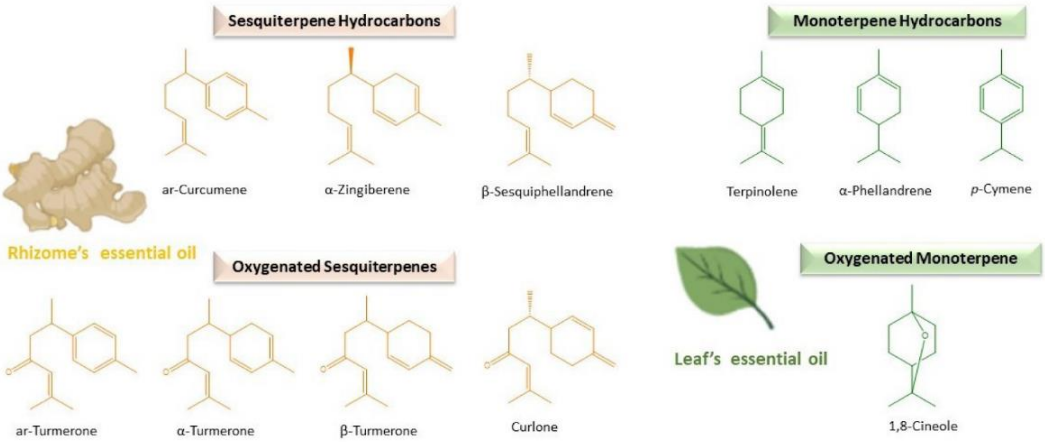


Figure 4: Chemical structure of Phytochemical compounds found in volatile oils from turmeric rhizome (Source: Ibáñez et al. 2021)

There are three types of curcuminoids found in turmeric that are accountable for the yellow color of the rhizome of turmeric - curcumin(diferuloylmethane), and its derivatives bisdemethoxycurcumin and demethoxycurcumin (Nisar et al. 2015) (Figure 5). The oleoresin, which is synthesized from rhizome extraction, contains 35-45% of

curcumin and its derivatives (https://uses.plantnet-project.org/e/index.php?title=Curcuma_longa). The organic extract of turmeric, oleoresin, can be used as a coloring agent for food items (Nair, 2019). Oleoresin also contains volatile oils (15-20%) besides curcuminoid pigments (Nair, 2019) (Table 2).

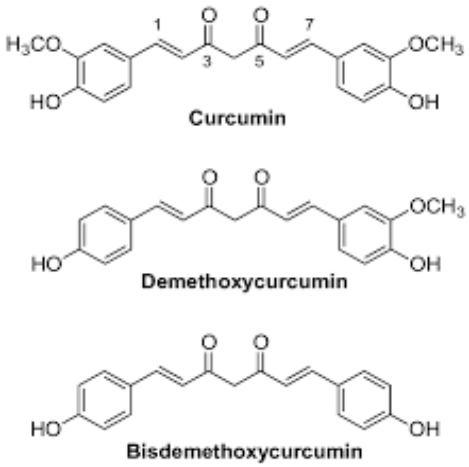


Figure 5: Chemical structures of curcumin and its derivatives (Source: https://www.intechopen.com/)

Table 2: Composition of Volatile Oils of Oleoresin

Compounds	Quantities in percentage (%)
Turmerone	60
Zingiberene	25
d-α-phellandrene	1
d-sabinene	0.6
Cineole	1
Farnesol	Small quantity

Source: Nisar et al. 2015 and Nair, 2019

The two major sesquiterpenes that are accountable for the unique fragrance of turmeric are ar-turmerone and turmerone (Nair, 2019). Depending on the maturity of rhizomes, volatile oil contains 24.7-48.9% ar-turmerone, and turmerone ranges from 20-39%.

The chemical composition of oleoresin of turmeric can fluctuate depending on the genetics of turmeric species and environmental factors (Ibáñez et al. 2021).

Medicinal properties of turmeric

In Ayurvedic and Unani (Ancient Indian medicinal practices), the fresh extract of turmeric rhizome has been used as an antiseptic, antioxidant, and anti-inflammatory for treating coughs, anorexia, and liver diseases (Nisar et al. 2015; Nair, 2019). Turmeric extract can be used for skin eruptions, smallpox, and to relieve eye sensation (plantnet-project.org). Fresh extract from turmeric rhizome is mixed with warm milk, which acts as a tonic for the common cold and a blood purifier (plantnet-project.org). The oleoresin from turmeric rhizome extract has curcumin which has an antioxidant, antibacterial, anti-inflammatory, and anticancer effect (Prasad et al. 2011). However, it has been proven that the fresh extract of rhizome has more antioxidant properties than dry rhizome powder (Nisar et al. 2015). So, the powder used as a spice in the Indian culinary world has minor antioxidant activities (Nair, 2019). Ar-turmerone from the volatile oil has anti-cancer and antibacterial potential, which prevents inflammatory diseases like psoriasis (Ibáñez et al. 2021). Researchers identified the derivatives of volatile oil extracted from *C. longa* rhizome, α-phellandrene, and ar-turmerone to have insecticidal properties, which are essential for agricultural pest issues (de Souza Tavares et al. 2016; Prasad et al. 2011). Turmeric has been a proven treatment in respiratory areas like bronchitis and asthma (Prasad et al. 2011).

For nearly 2000 years, ancient Chinese medicine has used turmeric to treat abdominal pain. Ancient Muslim medicinal practitioners (Unani) in India utilized turmeric rhizome

extract for a better blood circulatory system (Prasad et al. 2011). In folk medicine, turmeric has been used to enhance digestion, normalize menstruation, strengthen body energy, dissolve gallstones, and remedy gastrointestinal problems such as irritable bowel syndromes (IBS). Turmeric is also used as a wound cleanser, skin remedy, blood purifier, and to keep injurious microbes away.

In modern medicine, turmeric has been used for prevention and treatment of many common ailments (Prasad et al. 2011). In recent years, in-vitro studies have proven the antioxidant properties of the fresh extracts of *C. longa* rhizome (Tilak et al. 2004; Prasad et al. 2011). It reduces cytotoxicity in renal epithelial cells induced by hydrogen peroxide (Prasad et al. 2011). Also, in an in-vitro study, turmeric is proven to increase revertants against mutagenicity produced by carcinogens and mutagens (Soudamini et al. 1995 and Prasad et al. 2011). One piece of evidence suggests that curcuminoids of turmeric crude organic extract have anti-inflammatory properties which treat swelling or inflammation. When HL-60 cells (a human leukemia cell line) were stimulated by turmeric oil, it showed a decrease in the production of tumor necrosis factor (TNF)-α and prostaglandin E2 (PGE2), which decreases the tumor cell proliferation (Lantz et al. 2005). For cardiovascular disease, turmeric is one prescribed treatment. One recent study has explained the effect of turmeric on atherosclerosis, where curcumin helps to reduce fats for better blood circulation (Hasan et al. 2014). The anti-diabetic effects of turmeric in Type 2 diabetes were found by Na et al. 2014. This study suggested curcumin can lower the stored glucose in type 2 diabetic patients' blood. Ar-turmerone (one of the chemical components of turmeric) induces apoptosis in human leukemia cells by causing DNA fragmentation, which has shown anti-cancer activities of turmeric (Aratanechemuge et al.

2002; Prasad & Aggarwal, 2011). Also, in various cell lines, turmeric is reported for its inhibition effect against skin cancer, breast cancer, oral cancer, and stomach cancer by inhibiting mutation and carcinogens and inducing apoptosis in cancer cells (Prasad & Aggarwal, 2011).

For various diseases, turmeric has been tested in humans. One study result suggests its antimutagenic effect and potential chemoprevention use (Polasa et al. 1992). It was also found to relieve peptic ulcers (Phase II clinical trial) and to heal irregular bowel syndrome (Prasad & Aggarwal, 2011).

Different Curcuma species in India, Variations in Curcuma species

The genetic background of the Curcuma genus is extensive. Curcuma genus has over 100 species (Table 3), and 41 are found in India (Nair, 2019). India is known to have 41 species, 50 cultivars, and 20 varieties of *Curcuma* (Table 4). Other species are mostly scattered throughout Southeast Asia. India and Thailand have the Curcuma species with the highest diversity. These diversities are due to hybridization between the wild species and polyploidy (Jana et al. 2007). Polyploidy has an important role in plant evolution through speciation, depending on the ecology and geography (Weiss-Schneeweiss et al. 2013). Besides polyploidy, genome size plays a useful role in finding variations as well as ecological behavior of species in the same genus (Jana et al. 2007). However, in the Curcuma genus, the genome size is not well known. One study has hypothesized that curcumin content in different species differs depending on the genetic variation and hybridization between species (Hayakawa et al. 2011). They studied the evolution of chloroplast DNA (cpDNA) of Curcuma species. They found that *Curcuma longa* has

highest curcumin content, and with hybridization of the cpDNA, the species has a gradient degradation of the content from medium to low (Figure 6).

Table 3: Different Species of Curcuma Genus

1. Curcuma aeruginosa (pink and blue ginger) 2. Curcuma albicoma 3. Curcuma albi ora 4. Curcuma alismatifolia (summer tulip) 5. Curcuma amada (mango ginger) 6. Curcuma amarissima 7. Curcuma Americana 8. Curcuma angustifolia (tall hidden ginger) 9. Curcuma aromatica 10. Curcuma attenuata 11. Curcuma aurantiaca (rainbow curcuma) 12. Curcuma aurantiflora 13. Curcuma australasica (Cape York turmeric) 14. Curcuma bakeriana 15. Curcuma bicolor (candy corn) 16. Curcuma brog 17. Curcuma burttii 18. Curcuma caesia 19. Curcuma cannanorensis 20. Curcuma cannanorensis var. Lutea	21. Curcuma caulina 22. Curcuma careyana 23. Curcuma certothecca 24. Curcuma chuanezhe 25. Curcuma chuanhuangjiang 26. Curcuma chuanyujin 27. Curcuma cochinchinensis 28. Curcuma codonantha 29. Curcuma coerulea 30. Curcuma colorata 31. Curcuma comosa 32. Curcuma cordata (jewel of Thailand) 33. Curcuma cordifolia 34. Curcuma coriacea 35. Curcuma decipiens 36. Curcuma domestica (Emperor variegated) 37. Curcuma ecalcarata 38. Curcuma ecomata 39. Curcuma elata (giant plume) 40. Curcuma erubescens 41. Curcuma euchroma 42. Curcuma euclroma 43. Curcuma exigua 44. Curcuma ferruginea	45. Curcuma flaviflora (red fireball ginger) 46. Curcuma glans 47. Curcuma glaucophylla 48. Curcuma gracillima (chocolate zebra) 49. Curcuma grahamiana 50. Curcuma grandiflora 51. Curcuma haritha 52. Curcuma harmandii (emerald pagoda ginger) 53. Curcuma heyneana 54. Curcuma inodora (pink ginger) 55. Curcuma karnatakensis 56. Curcuma kudagensis 57. Curcuma kwangsiensis 58. Curcuma kwangsiensis var.affinis 59. Curcuma kwangsiensis var. puberula 60. Curcuma lanceolata 61. Curcuma latiflora 62. Curcuma latifolia 63. Curcuma leopoldi 64. Curcuma leucorhiza 65. Curcuma loerzingii 66. Curcuma lillicina (pink cloud)	67. Curcuma longa (turmeric) 68. Curcuma longiflora 69. Curcuma longi spica 70. Curcuma lutea 71. Curcuma malabarica 72. Curcuma mangga 73. Curcuma meraukensis 74. Curcuma montana 75. Curcuma musacea 76. Curcuma mutabilis 77. Curcuma neilgherrensis 78. Curcuma nilamburensis 79. Curcuma ochrorhiza 80. Curcuma of cinalis 81. Curcuma olena 82. Curcuma oligantha (white turmeric) 83. Curcuma oligantha var. lutea 84. Curcuma ornate (ornate plume ginger) 85. Curcuma pallida 86. Curcuma parviflora (white angel) 87. Curcuma parvula 88. Curcuma peethapushpa	89. Curcuma petiolata (Temu Puteri in Java) 90. Curcuma phaeocaulis 91. Curcuma pierreana (sleeping princess) 92. Curcuma plicata 93. Curcuma porphyrotaenia 94. Curcuma prakasha 95. Curcuma pseudomontana 96. Curcuma purpurascens 97. Curcuma purpurea 98. Curcuma raktakanta 99. Curcuma ranadei 100. Curcuma reclinata 101. Curcuma rhabdota 102. Curcuma rhomba 103. Curcuma roscoeana (pride of Burma) 104. Curcuma rotunda 105. Curcuma rubescens (wine red plume) 106. Curcuma rubricaulis 107. Curcuma rubrobracteata (fire plug) 108. Curcuma sessilis 109. Curcuma siamensis 110. Curcuma sichuanensis 111. Curcuma singularis (Easan white)

Source: Prasad et al. 2011

Table 4: Different species of Curcuma found in India (Source: Nair, 2019)

Species	Statewise geographic distribution
C. aeruginosa	West Bengal
C. albiflora	Kerala
C. amada	Throughout India
C. amarissima	West Bengal
C. angustifolia	Uttar Pradesh, Madhya Pradesh, Himachal Pradesh, Northeast India
C. aromatica	Kerala, Tamil Nadu, Karnataka, Andhra Pradesh, Odisha, and Bihar
C. caesia	West Bengal
C. caulina	Maharashtra
C. comosa	West Bengal
C. petiolata	West Bengal
C. decipiens	Kerala and Karnataka
C. rubescens	West Bengal
C. ferruginea	West Bengal
C. longa	Throughout India
C. montana	Kerala, Karnataka, Tamil Nadu, and Andhra Pradesh
C. neilgherrensis	Same as above
C. oligantha	Kerala
C. pseudomontana	Kerala, Karnataka, Tamil Nadu, and Andhra Pradesh
C. reclinata	Madhya Pradesh
C. xanthorrhiza	West Bengal
C. zedoaria	Throughout India
C. sylvatica	Kerala
C. aurantiaca	Kerala, Karnataka, Tamil Nadu, and Andhra Pradesh
C. sulcata	Maharashtra
C. indora	Gujarat, Maharashtra, and Karnataka
C. ecalcarata	Kerala
C. soloensis	West Bengal
C. brog	West Bengal
C. haritha	Kerala
C. raktakanta	Kerala
C. kudagensis	Karnataka
C. thalakaveriensis	Karnataka
C. malabarica	Kerala and Karnataka
C. karnatakensis	Karnataka
C. cannanorensis	Kerala
C. vamana	Kerala
C. lutea	Kerala and Karnataka
C. coriacea	Kerala
C. nilamburensis	Kerala
C. leucorhiza	West Bengal

Source: Velayudhan et al. (1999)

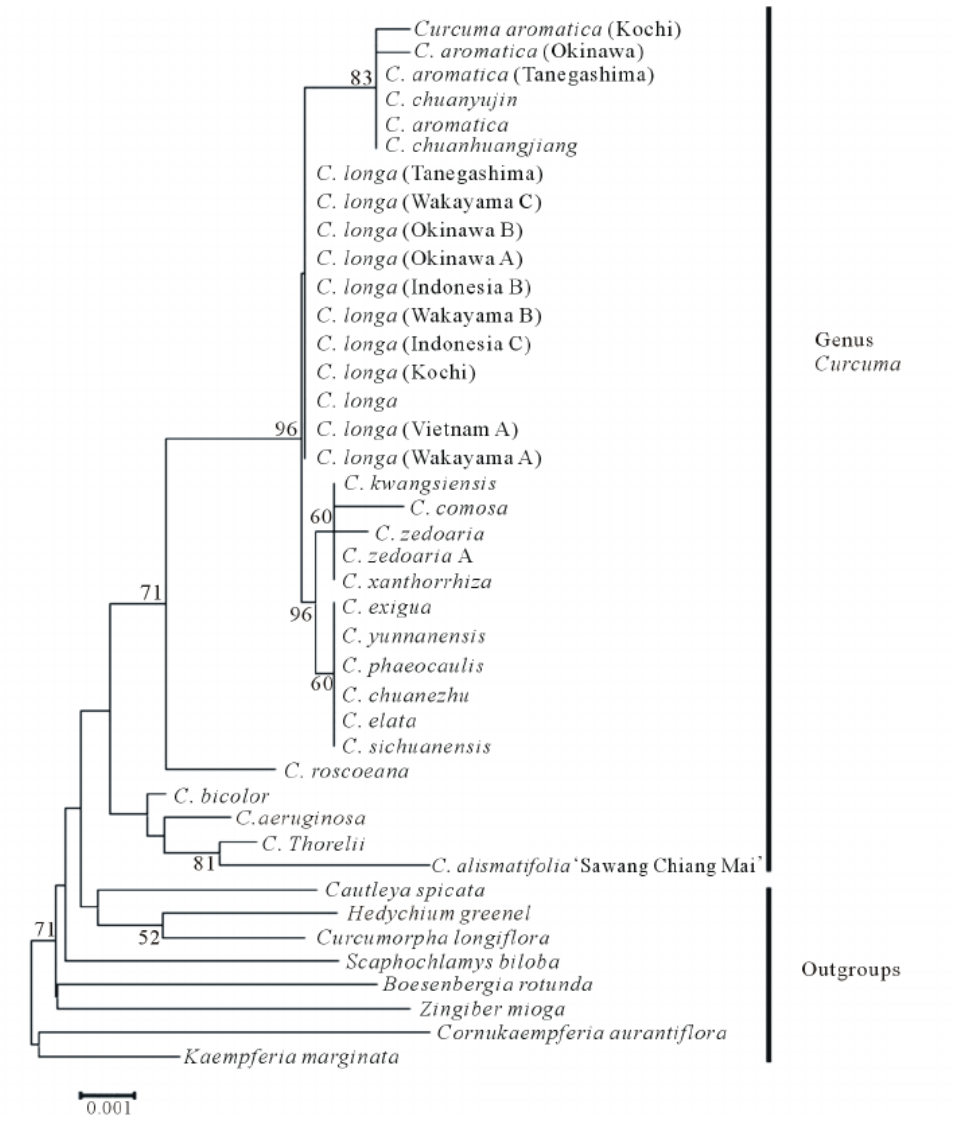


Figure 6: Phylogenetic tree of Curcuma and its different species based on Chloroplast genome (cpDNA) using neighbor-joining method (Hayakawa et al. 2011)

From the phylogenetic tree studies based on the genomic DNA with bootstrap support, it is found that *Curcuma longa* L. has a close relationship with *Zingiber officinale* (Sass et al. 2016) (Figure 7).Ginger has been studied b roadlyby researchers at molecular level. The reference genome of ginger has already been submitted to NCBI by Ori-Gene Co. Ltd. (RefSeq: GCF_018446385.1, GenBank: GCA_018446385.1).

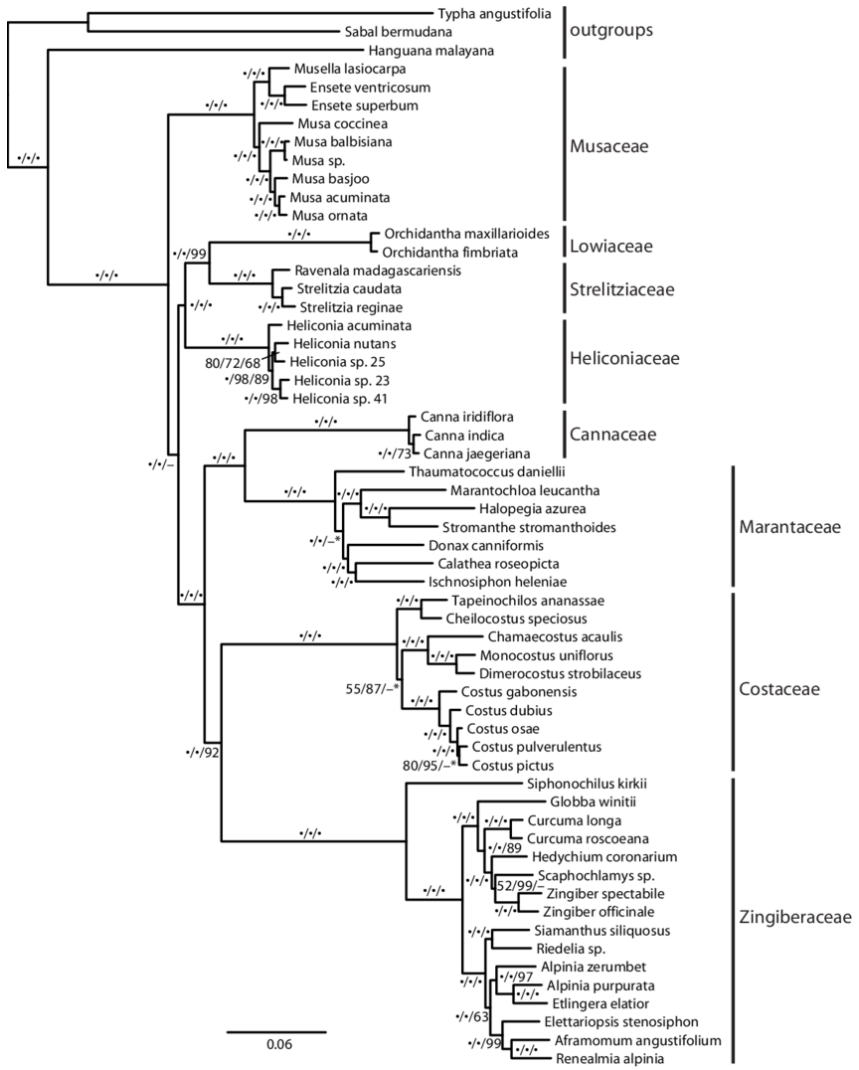


Figure 7: Phylogenetic tree of Zingibarels family based on the Maximum Likelihood (ML) method of genomic DNA (Source: Sass et al. 2016)

Cultivation of turmeric

Turmeric is mainly cultivated in tropical areas (Figure 8). In Southern Asia, turmeric is most extensively grown in: Southern China, Taiwan, Japan, Burma, and Indonesia, including India and Bangladesh, as well as in the African continent (Tanvir et al. 2017). Turmeric production in India is 225000 metric tons per year, about 94% of the world's turmeric production. Most of the production is used as a powder for coloring as well as a

spice. The United States imports about 2043 metric tons of turmeric every year; most of it originated in India (Buescher & Yang, 2000). In India, Assam, Karnataka, Tamil Nadu, and West Bengal are the top producers but all these areas are experiencing effects of climate change.

Figure 8: Cultivation of turmeric (Source: krishijagran.com)

Drought stress

In agriculture, often unfavorable environmental conditions are faced by plants all the time due to drastic global climate changes. Drought is one of the critical abiotic stresses that causes most of the losses in the crop fields (Santhi et al, 2016). Drought stress can change a plant at morphological, anatomical, physiological, biochemical, and molecular levels (Seleiman et al, 2018). These alterations in plants are mostly due to adapting to the unwanted environmental stress. Various studies have been reported to date to address the adverse effects of drought stress on plants as well as how to mitigate the negative effect by improving the genetics of plants with the help of biotechnology and genetic engineering. To do so, it is essential to know how the plant responds towards a specific abiotic stress factor. Drought stress effect in plants can primarily be identified by

decreased photosynthesis, chlorophyll content, moisture content, stomatal closure, and carbohydrate metabolism. Photosynthesis is one of the major mechanisms to maintain plant growth and development. Drought stress reduces the efficiency of the photosynthetic apparatuses, such as the thylakoid membrane, which gets damaged and eventually reduces the chlorophyll content (Song et al. 2021). The chloroplast is the place where the dark and light reactions of photosynthesis take place (Muhammad et al. 2021). These chloroplasts are highly responsive towards drought stresses (Muhammad et al. 2021 & Song et al. 2021). Under severe abiotic stress, the chloroplast generates a series of Reactive Oxygen Species (ROS) such as superoxide, hydrogen peroxide, hydroxyl radical and singlet oxygen (Song et al. 2021& Hussain et al. 2018). However, in the earlier stage of drought stress, the plant provides a responsive mechanism to reduce the overproduction of ROS, which is the production of enzymatic and non-enzymatic antioxidants. These enzymatic oxidants (such as catalase, superoxide dismutase, etc.) and non-enzymatic antioxidants (such as ascorbic acid, carotenoids, etc.) scavenge the ROS to provide tolerance to plants against drought stress. Many studies also reported that many genes have been identified as drought stress-specific genes (Hussain et al. 2018). Under mild drought stress, plants control net CO_2 and transpiration flow by inhibiting photosynthesis and stomatal conductance. However, under severe drought stress, water use efficiency (WUE) and root conductivity decreases due to the inhibition of the metabolic mechanism of photosynthesis (Song et al. 2021). Also, the inhibition of the electron transport system under drought stress can induce an alternative oxidase pathway (a mitochondrial metabolic electron transport chain) which can be essential to reduce the production of ROS and to maintain the metabolites under stress (Vanlerberghe, 2013).

Plant growth regulators also plays an essential role in drought tolerance in plants. Under drought stress, the abscisic acid (ABA) biosynthesis increases (Zhang & Davies, 1987). These increased ABA regulates an interacting hormonal network between cytokinin, ethylene and auxin to maintain root growth under drought stress (Rowe et al. 2016). Auxins induction during drought stress provides photosynthesis stability and normal stomatal conductance (Song et al. 2021). The synthesis of ethylene can also provide drought tolerance to plants (Song et al. 2021).

Usually, turmeric prefers moist soils in a well-shaded area. It usually does not tolerate dried-out soil. Because of that, tropical regions like India face challenges while growing turmeric during summer (https://deepgreenpermaculture.com/2019/02/10/growing-turmeric-in-the-ground-in-temperate-climates/). In a case study in 2018, it was reported that due to lack of rains, turmeric lands are facing drought, which threatens to hit turmeric production (http://www.commoditiescontrol.com/eagritrader/common/). In the states of Maharashtra, Tamil Nadu, Telangana &Andhra Pradesh, in India, during a drought in 2018, the production of turmeric declined, and there was not enough supply in accordance with the demand (commoditiescontrol.com, 2018).

Therefore, it is important to understand the effect of drought stress on turmeric at the morphological, anatomical, physiological, and molecular level. If we get to know the changes at the molecular level, and its functional annotation of transcriptomics, then it maybe possible to genetically improve drought-tolerant turmeric plants to get better production and to meet the increasing demand in the future.

Chapter 2

Materials and Methods

Turmeric rhizomes (CL11 strain) which were supplied by Alabama A&M University, were used for this study and grown in the greenhouse of California State University, Northridge, till they sprouted and later moved to a temperature (24-26 °C) and light-controlled (light cycle:16/8) growth chamber. CL11 strain rhizomes were planted in pots. After two months, they grew up into mature plants. Watering is withdrawn from three pots for 21 days to mimic drought stress. And three pots were selected as controls by watering them every alternative day. After 21 days, leaves and rhizomes are collected, and physiological tests were performed on the fresh leaves. Some of the samples were stored at -80°C for further future analysis.

Physiological studies

For all physiological studies, two drought-stressed plants were used as the third plant did not have enough leaves to experiment with both molecular and physiological studies.

Photosynthetic rate

All the photosynthetic phenotypes were measured by using a handheld instrument called MultiSpeQ (Figure 9). The metadata collected using MultispeQ gets saved in an online database platform called PhotosynQ platform (http://photosynq.org). The platform provided web-based tools to visualize and analyze the tools. (Kuhlgert et al. 2016).

Figure 9: MultiSpeQ meter to scan plant material

The measures were taken at three different points in each biological replicate (Three controlled plants and three stressed plants) (Ben-Jabeur et al. 2021). The measurements taken by MultispeQ were total non-photochemical quenching (NPQt), quantum yield of PSII (Phi2), fraction of light lost via nonregulated photosynthesis inhibitor processes (PhiNO), fraction of light dedicated to non-photochemical quenching (PhiNPQ), fraction of PSII open centers (qL), total electrochromic shift (ECSt), maximal quantum efficiency of PSII (Fv′/Fm′), Leaf temperature, Leaf thickness, linear electron flow (LEF).

The effect of water shortage on photosynthetic rate was analyzed through one factor ANOVA using RStudio 4.2.0 (https://www.rstudio.com/).

Chlorophyll content

For the following method, only two stressed biological replicates (S1 &S3) were used as the Sample 2 plant did not have enough leaves.

It was estimated by following the protocol given by Hiscox and Israelstom (1979). Freshly collected turmeric leaves from each biological sample were weighed. 0.5 g of leaves were cut into small pieces and set in test tubes with 5 mL of DMSO and afterward

warmed in a water bath for 2.5 hours at 50°C. The absorbance of each yellow or green supernatant was assessed at 645 nm and 663 nm and blanked utilizing DMSO on a NanoDrop UV Visible Spectrophotometer (Thermo Scientific, USA). In conclusion, complete chlorophyll content was determined by the equation given by Arnon (1949). These included:

Chlorophyll a (mg/ ml) = $(12.7 \times Abs_{663\ nm}) - (2.69 \times Abs_{645\ nm})$

Chlorophyll b (mg/ ml) = $(22.9 \times Abs_{645\ nm}) - (4.68 \times Abs_{663\ nm})$

Total chlorophyll (mg/ ml) = chlorophyll a + chlorophyll b

Relative Moisture content

Relative Moisture Content (RMC) was assessed by following the method given by Singh and Rao, 1983. Three leaves (technical replicates) from each biological replicate (three control plants and two drought-stressed plants) were collected and cut into small pieces. Ten symmetrical pieces from each sample were collected and weighed (fresh weight). Then they were submerged in distilled water for four hours and weighed again (submerged weight). After weighing, they were dried for 14 hours in an incubator set to 43°C-45°C and weighed once more (dried weight)

Ultimately, RMC was determined using the following formula:

(Fresh weight - Dry weight)/ (Submerged weight - Dry weight)

Statistical analysis: The effect of water shortage on RMC was analyzed through one factor ANOVA using RStudio 4.2.0 (https://www.rstudio.com/).

Catalase activity

Freshly gathered leaves from the three control plants and two drought-stressed plants (4.0 g of leaves from each biological replicates) were ground in 8 ml of distilled water. Then 2 ml of the aqueous solution was pipetted into a 50 mL graduated cylinder, and 2 mL of 3% H_2O_2 was added. Finally, catalase activity was calculated by the volume of effervescence as given by Euler (1927).

Molecular studies

Total RNA extraction

It can be challenging to isolate total RNA from plant cells as it has polyphenols and polysaccharides. Plants like turmeric have a lot of secondary metabolites, which makes them more complicated. Pure RNA isolation was confirmed by using Nanodrop Spectrophotometer and RNA formaldehyde-free gel electrophoresis.

High-quality RNA is preferred for further processing in gene expression analysis. Purified RNA is free of genomic DNA, phenols, polysaccharides, and secondary metabolites. For RNAseq, genomic DNA contamination can affect analysis as the sequencing reaction in the flow cell cannot differentiate between cDNA and DNA contamination.

In this study, high-quality RNA was extracted using Trizol (Thermo Fisher, USA) modified by Basu's lab (Ramadoss &Basu, 2018). RNA quality was improved by using an RNA clean and concentrator kit (Zymo Research, USA) with some modifications from Basu's lab (Ramadoss & Basu, 2018).

Before using, all the equipment (including gloves) and all materials were sprayed with RNase (Ribonuclease) AWAY (Thermo Fisher, USA). Approximately 100 mg of frozen leaves are weighed and snap frozen again by immersing them in liquid nitrogen for five seconds. RNase AWAY sprayed mortar and pestle were used to grind the frozen leaves in the presence of 1ml of Trizol. Different sets of mortar pestles were used to avoid cross-contamination between the samples. Ground samples were then transferred to a nuclease-free microcentrifuge tube using a pipette. Then, the tube was incubated at room temperature for five minutes. Then, 200 µl of 24:1 Chloroform: Isoamyl alcohol was added and vortexed vigorously to mix thoroughly. The vertexing was continued till the mix was homogeneously green. This step was performed carefully as it is critical for a higher yield of RNA extraction. The mixture was incubated at room temperature for two minutes and centrifuged for fifteen minutes at 4ºC at 12000×g. After the centrifugation, we saw three phases, the clear aqueous phase contained RNA. So, the aqueous phase was transferred carefully to a new tube. Then, 0.5 times the aqueous phase volume of chilled isopropanol (must be chilled, otherwise the probability of low RNA yield will be higher) was added to the tube. Again, vortexed vigorously for ten seconds. After fifteen minutes of incubation at -20ºC, the tube was centrifuged for ten minutes at 12000g at 4ºC. At this point, RNA was seen as a pale white pellet at the bottom of the microcentrifuge tube. Once we saw the RNA, the supernatant was carefully discarded, and the pellet was washed by adding 500 µl of 75% chilled ethanol. Carefully the ethanol was again discarded after the centrifugation at 7000 g for five minutes. The washed tube was again spun with an open cap (without adding any reagents) to remove all additional ethanol. Tubes were then air dried for fifteen minutes by placing them invertedly with their caps

open. Air-dried pellets in the tube were then resuspended in 40 µl of nuclease-free water and then incubated for fifteen minutes at 60ºC in a water bath. While doing this step, the cap of the tube was open to evaporate all the residual ethanol that had been used for the RNA extraction. In this step, finally, isolated RNA can be seen as a pellet; however, they are not purified. So, some extra steps to get RNA in a purified form were performed.

RNA purification

The protocol given by Ramadoss & Basu, 2018 was followed and modified from ZYMO Clean and Concentrator KitTM protocol (Zymo Research, USA).

First, 80 µl of RNA binding buffer was added to the 40 µl isolated RNA and gently mixed in the solution by pipetting up and down ten times. Then, 120 µl of chilled 100% ethanol was added, followed by gentle mixing, and transferred to a Zymospin column in a collection tube. The tube was centrifuged for 30 seconds at 12,000 g at 4ºC. A washing step was followed right after by adding 400 µl of RNA wash buffer and spinning at 12000×g for 30 seconds at 4ºC, and the flow through was discarded. After this step, 40 µl of DNase cocktail was added to the RNA to remove all DNA contamination from the solution. The DNase cocktail was prepared by gently mixing 5µl of Turbo DNase (Thermo Fisher Scientific, AmbionTM) and 35µl of DNA digestion buffer (ZYMO RESEARCH, USA) in a microfuge tube. Incubation can be done for up to 20 minutes at room temperature. After the incubation, 400µl of RNA prep buffer was added to the Zymo column (Zymo Research, USA), and the flow through was discarded after spinning the column at 12000×g for 30 seconds at 4ºC. The washing step was repeated using 400 µl of RNA wash buffer and discarded the flow through. An extra spinning of the column

was followed without adding any reagents after the second washing step to remove all the residual ethanol. Lastly, the column was transferred to a new nuclease-free microcentrifuge tube, and 20 µl of nuclease-free water was added to the column. Incubation at room temperature for two minutes was required, and after that, the tube was spun at 12000×g for two minutes at 4ºC. Then, the flow through was collected and again added to the column. After incubation for two minutes at room temperature, the tube was centrifuged for two minutes at 12000×g at 4ºC, and then the pure RNA was collected

RNAseq library preparation

The RNAseq library was prepared using the Truseq RNA library prep kit v2 LS protocol (Illumina, Inc). This protocol can only be used when the total sample is less than 48. Sample preparation is a long and continuous process. There are steps between which can take a maximum of 7 days with breaks.

I) Preparation of purified and fragmented mRNA

Before starting, it is essential to check all reagents are ready. After checking all the required reagents were thawed at room temperature.

Thermocycler was pre-programmed with the following:

mRNA Denaturation: The thermocycler was set at 100°C by choosing a pre-heat lid, followed up with 65°C for 5 minutes, and then held at 4°C

mRNA Elution 1: The thermocycler was set at 100°C by choosing a pre-heat lid, 80°C for 2 minutes, and then held at 25°C.

Elution 2 - Frag – Prime: The thermocycler was set at 100°C by choosing a pre-heat lid, then 94°C for 8 minutes, and held at 4°C.

The centrifuge was set at room temperature.

1.Preparation of RNA bead plate (RBP) plate

400 ng per µl of pure total RNA was used to prepare the library.

All total RNA samples were loaded on the 0.3 ml PCR plate labeled as the RBP plate, then diluted to a final volume of 50 µl with nuclease-free ultrapure water. [Resuspension of the room temperature RNA purification beads was completed by vortexing it vigorously]. 50 µl of RNA purification beads were added to each well of the RBP plate. (This step was performed so that poly-A tails of mRNA bind to the oligo-dT magnetic beads). Next, the mixture was gently pipetted up and down the entire volume six times. Lastly, the plate was sealed with Microseal`B´Adhesive Seals.

2.Incubation of RBP plate 1

The sealed RBP plate was placed on the thermocycler which facilitated the binding of the poly-A tails to the beads. Then, the plate was removed from the thermocycler when it reached 4 °C. After removal, the plate was placed on the bench at room temperature before the next step. T

3.Washing of RBP Plate 1

After the room temperature incubation, the seal was removed from the plate and placed on a magnetic stand for five minutes at room temperature. This step allows the separation of poly-A bound RNA beads from the rest of the solution. Then the supernatant was discarded from each well. It was easy to distinguish the pellet (poly-A bound RNA beads) from the supernatant (the solution). Then, the plate was removed from the magnetic stand. The pellet was then washed by adding 200

µl of bead washing buffer in each well of the plate. The mixture was gently pipetted up and down six times to mix it thoroughly. This step is helpful in removing all the unbounded RNA; after mixing, the RBP plate was again placed on the magnetic stand for five minutes at room temperature. Then, the supernatant was discarded. The supernatant mostly contains all ribosomal and other non-messenger RNA. Then, the plate was removed from the magnetic stand and 50 µl of elution buffer was added to each well of the plate. (The thawed elution buffer was centrifuged for five seconds at 600×g). Elution buffer was well mixed into the solution by pipetting up and down 6 times. Then the RBP plate was sealed.

4.Incubation of RBP plate 1

The sealed RBP plate was placed on the thermocycler and the program for the mRNA Elution 1 was run. This step facilitated the elutes of mRNA from the beads and reduced the probability of non-specific bounds. Then, the plate was removed from the thermocycler when it reached 25 ℃. After removal, the plate was placed on the bench at room temperature before proceeding to the next step. This incubation helps the bead to bind to the RNA. After that the seal was removed from the plate.

5.Preparation of RBP plate 2

The thawed Bead Binding Buffer (BBB) was centrifuged for five seconds at 600×g. Then 50 µl of BBB was added to each well of the RBP plate. This step again facilitated the specific binding of mRNA to the beads and reduced the chances of non-specific binding of rRNA and others. After adding, the whole solution was gently pipetted up and down six times to mix it thoroughly. Then,

the plate was placed on the bench for incubation at room temperature for five minutes. After incubation, the RBP plate was placed on a magnetic stand for five minutes at room temperature. After that, all supernatants were discarded, and then, the plate was removed from the magnetic stand. The pellet was then washed by adding 200 µl of bead washing buffer to each well of the plate. The mixture was gently pipetted up and down six times to mix it thoroughly. This step is helpful in removing all the unbounded RNA, after mixing, the RBP plate was placed on the magnetic stand for five minutes at room temperature. Then, the supernatant was discarded. The supernatant mostly contains all ribosomal and other non-messenger RNA. Then, the plate was removed from the magnetic stand, and 19.5 µl of Elute, Prime, and Fragment Mix was added to each well of the plate. Elute, Prime, and Fragment Mix was well mixed to the solution by pipetting up and down six times. This mix contains random hexamers for RT priming and acts as the 1st strand cDNA synthesis reaction buffer. Then the RBP plate was sealed as above.

6.Incubation of RBP plate 2

The sealed RBP plate was placed on the thermocycler and the program named the Elution-2-Frag-Prime was run. This step facilitated the elutes, fragmentation of RNA. Then, the plate was removed from the thermocycler when it reached 4°C. After removal, the plate was centrifuged for 1 min at 280 g.

II) Preparation of first strand cDNA synthesis

This step uses reverse transcriptase and reverse primers for reverse transcription of cleaved RNA fragments primed with random hexamers to synthesize first strand cDNA.

Before starting, it is important to check all reagents are ready. After checking all the required reagents were thawed at room temperature.

Thermocycler was pre-programmed with the following:

1st strand: The thermocycler was set at 100 °C by choosing pre-heat lid, followed up with 25°C for 10 minutes, 42°C for 50 minutes, 70°C for 15 minutes, and then held at 4°C.

1.Preparation of cDNA plate (CDP) plate

RBP plates were placed on the magnetic stand at room temperature for five minutes. While placed on the stand, the adhesive seal was removed and 17 µl of the supernatant was transferred from each well of RBP plates to the corresponding new well of the new 0.3 ml PCR plate which was labeled as CDP. The first Strand Master Mix tube was centrifuged for five seconds at 600×g. SuperscriptII was added to the mix with a ratio of 1:9. Six µl of Superscript II was added to 54 µl of First Strand Master Mix. A total of 60 µl of mix was prepared. Then 8 µl of the first strand mix was added to each well of the CDP plate and then pipetted up and down the whole mix six times to mix it thoroughly. Then the CDP plate was sealed as above.

2.Incubation of CDP plate 1

The sealed CDP was placed on the thermocycler and the program named the 1st strand was run. Then, the plate was removed from the thermocycler when it reached 4°C.

III) Preparation of Second strand cDNA synthesis

Prior to use, the AMPure beads were incubated at room temperature nad then vortexed until they were well-dispersed, and the color appeared homogeneous. Beads were not reused.

1.Adding the Second Strand Master Mix (SSM)

First, SSM was centrifuged for five seconds at 600×g. Then, the adhesive seal was removed from the CDP plate and 25 µl of SSM was added to each well of the plate. The mix was gently pipetted up and down six times to mix it thoroughly. Then, the CDP plate was sealed as above.

2.Incubation of CDP plate 2

The CDP plate was placed on the pre-heated thermal cycler and was incubated for one hour at 16°C. Then, the plate was removed from the thermal cycler and the seal was removed. Plate was placed on the bench to incubate at room temperature.

3.Purification of CDP plate

90 µl of the AMPure XP beads were added to each well of the CDP plate which already contained 50 µl ds cDNA. The mix was gently pipetted up and down six times to mix it thoroughly. Then, it was incubated at room temperature for 15 minutes. Then the plate was placed on the magnetic stand at room temperature for five minutes. 135 µl of supernatant was then removed and discarded from each well of the plate. While on the magnetic stand, each well of the CDP plate was filled with 200 µl of freshly prepared 80% ethanol without touching the beads. CDP plate was then again incubated at room temperature for 30 seconds and then all supernatants were discarded. This was repeated once again for a total of two ethanol washes. After the wash, the plate was incubated at room temperature for

15 minutes to dry and then it was removed from the magnetic stand. Prior to use, the resuspension buffer was centrifuged for five seconds at 600×g and then 52.5 µl of it was added to each well of the CDP plate. The mix was gently pipetted up and down ten times to mix it thoroughly. Then incubation was done at room temperature for two minutes. After incubation, the plate was placed on the magnetic stand for 5 minutes at room temperature. Then, 50 µl of the supernatant from each well of the CDP plate was transferred to new corresponding well of the new 0.3 ml PCR plate labeled as IMP plate.

After this step, the IMP plate was stored with the adhesive seal at -20°C for 24 hours and then proceeded to the next step.

IV) Performing end repair

This step converted the overhangs to blunt ends by using an end repair mix. The mix contains 3′ to 5′ exonuclease which removes the 3′ overhangs and polymerase which adds nucleotide to 5′ overhangs.

Thermocycler was pre-programmed with setting them at 30°C by choosing pre-heat lid

The of insert modification plate (IMP) plate was removed from -20°C and incubated at room temperature. Prior to use, centrifuged the thawed plate at 280 g for one minute.

1.Preparation IMP plate

10 µl of resuspension buffer was added to each well of the IMP plate that was contained with 50 µl ds cDNA. Then 40 µl of end repair mix was added. The mix was gently pipetted up and down ten times to mix it thoroughly. Then, the IMP plate was sealed as above.

2.Incubation of IMP plate 1

The sealed IMP plate was placed on a preheated thermocycler. After closing the lid, the plate was incubated for 30 minutes at 30°C. After incubation, the plate was removed from the thermal cycler.

3.Cleaning up the IMP plates

The adhesive seal was removed from the IMP plate. 160 µl pf well-dispersed AMPure XP beads were added to each well of the IMP plate which contained 100 µl of End Repair Mix. The mix was gently pipetted up and down ten times to mix it thoroughly. The mix was then incubated at room temperature for 15 minutes. The IMP plate was then placed on the magnetic stand for five minutes at room temperature. Then 127.5 µl supernatant was removed from each well of the plate and then discarded. Another 127.5 µl was again removed and discarded. While on the magnetic stand, each well of the IMP plate was filled with 200 µl of freshly prepared 80% ethanol without touching the beads. The IMP plate was then again incubated at room temperature for 30 seconds and then all supernatants were discarded. This was repeated once again for a total of two ethanol washes. After the wash, the plate was incubated at room temperature for 15 minutes to dry and then it was removed from the magnetic stand. Prior to use, the resuspension buffer was centrifuged for 5 seconds at 600×g and then 17.5 µl of it was added to each well of the IMP plate. The mix was gently pipetted up and down 10 times to mix it thoroughly. Then incubation was done at room temperature for 2 minutes. After incubation, the plate was placed on the magnetic stand for 5 minutes at room temperature. Then, 15 µl of the supernatant from each well of the IMP plate was

transferred to the corresponding well of the new 0.3 ml PCR plate labeled as adapter ligation plate (ALP).

After this step, the ALP plate with the adhesive seal was stored at -20°C for 5 days and then proceeded to the next step.

V) Adenylation of 3′ ends

A ′T′ nucleotide on the 3′ end of the blunt fragments was added in this step to prevent ligation to one another during the adapter reaction mix.

Thermocycler was pre-programmed with the following:

ATAIL70: The thermocycler was set at 100 °C by choosing pre-heat lid, which was followed by 37°C for 10 minutes 70°C for 5 minutes, and then held at 4°C.

The ALP plate was removed from -20°C and incubated at room temperature. Prior to use, centrifuged the thawed plate at 280×g for 1 minute. The adhesive seal was removed.

1.Addition of A-Tailing Mix (ATL)

2.5 µl of resuspension buffer was added to each well of the ALP plate. Then 12.5 µl of A-Tailing Mix (ATL) was added. The mix was gently pipetted up and down 10 times to mix it thoroughly. Then, the ALP plate was sealed as above.

2.Incubation of ALP plate 1

The sealed ALP plate was placed on a preheated thermocycler and the program named ATAIL70 was run. After incubation, the plate was removed from the thermal cycler. Then, the plate was removed from the thermocycler when it reached 4°C. After removal, immediately the next step was proceeded.

VI) Ligation of adapters

This step ligated multiple indexing adapters to the ends of ds cDNA for the flow cell sequencing.

6 adapters were used for 6 samples (Table 5).

Table 5: Adapter Index name and sequences

Samples	Adapter Index	Adapter sequences
C1 (Control 1)	2	CGATGT
C2 (Control 2)	7	CAGATC
C3 (Control 3)	19	GTGAAA
S1 (Stressed 1)	5	ACAGTG
S2 (Stressed 2)	6	GCCAAT
S3 (Stressed 3)	15	ATGTCA

Thermocycler was pre-programmed by setting at 30 °C by choosing pre-heat lid. The The 1st 96-well PCR plate was labeled as clean up ALP plate (CAP plate). The 2nd 96-well PCR plate was labeled as polymerase chain reaction plate (PCR plate).

1.Addition of ligation mix

The required adapter index tubes and stop ligation buffer tubes were centrifuged at 600×g for 5 seconds. Prior to use only, the ligation tube mix was removed from -20°C. The adhesive seal was removed from the ALP plate. 2.5 µl of resuspension buffer was added to each well of the ALP plate. Then 2.5 µl of Ligation mix was added. Then 2.5 µl RNA adapter indexes (as mentioned in the table 5) were added

to each well of the ALP plate. The mix was gently pipetted up and down 10 times to mix it thoroughly and then, the ALP plate was sealed as above.

2. Incubation of ALP plate 2

The sealed ALP plate was incubated on a preheated thermocycler for 10 minutes at 30°C. After incubation, the plate was removed from the thermal cycler.

3.Addition of stop ligation buffer (STL)

The adhesive seal was removed from the plate and the 5 µl of Stop Ligation Buffer was added to each well to inactivate the ligation. The entire volume was gently pipetted up and down 10 times to mix it thoroughly.

4.Cleaning up the ALP plates

42 µl well-dispersed AMPure XP beads were added to each well of the ALP plate. The mix was gently pipetted up and down 10 times to mix it thoroughly. The mix was then incubated at room temperature for 15 minutes. The ALP plate was then placed on the magnetic stand for 5 minutes at room temperature. Then 79.5 µl supernatant was removed from each well of the plate and then discarded. While on the magnetic stand, each well of the ALP plate was filled with 200 µl of freshly prepared 80% ethanol without touching the beads. The ALP plate was then again incubated at room temperature for 30 seconds and then all supernatants were discarded. This was repeated once again for a total of two ethanol washes. After the wash, the plate was incubated at room temperature for 15 minutes to dry and then it was removed from the magnetic stand. Prior to use, the resuspension buffer was centrifuged for 5 seconds at 600×g and then 52.5 µl of it was added to each well of the ALP plate. The mix was gently pipetted up and down 10 times to

mix it thoroughly. Then incubation was done at room temperature for 2 minutes. After incubation, the plate was placed on the magnetic stand for 5 minutes at room temperature. Then, 50 µl of the supernatant from each well of the ALP plate was transferred to the corresponding well of the new 0.3ml PCR plate labeled as Cleaning up ALP plate (CAP).

5.Second Cleaning up the CAP plates

50 µl of the AMPure XP beads were added to each well of the CAP plate. The mix was gently pipetted up and down 10 times to mix it thoroughly. The mix was then incubated at room temperature for 15 minutes. The ALP plate was then placed on the magnetic stand for 5 minutes at room temperature. Then 95 µl supernatant was removed from each well of the plate and then discarded. While on the magnetic stand, each well of the CAP plate was filled with 200 µl of freshly prepared 80% ethanol without touching the beads. The CAP plate was then again incubated at room temperature for 30 seconds and then all supernatants were discarded. This was repeated once again for a total of two ethanol washes. After the wash, the plate was incubated at room temperature for 15 minutes to dry and then it was removed from the magnetic stand. Prior to use, the resuspension buffer was centrifuged for 5 seconds at 600×g and then 22.5 µl of it was added to each well of the CAP plate. The mix was gently pipetted up and down 10 times to mix it thoroughly. Then incubation was done at room temperature for 2 minutes. After incubation, the plate was placed on the magnetic stand for 5 minutes at room temperature. Then, 20 µl of the supernatant from each well of the CAP plate

was transferred to the corresponding well of the new 0.3ml PCR plate labeled as PCR plate.

After this step, the CAP plate with the adhesive seal was stored at -20℃ for 24 hours and then proceeded to the next step.

VII) PCR amplification for enrichment of DNA fragments

This step was performed to amplify and enrich the DNA fragments which have adapter indexes on both ends.

Thermocycler was pre-programmed with the following:

PCR: The thermocycler was set at 100 ℃ by choosing pre-heat lid which was followed up with 98℃ for 10 seconds. 15 cycles of 98℃ for 10 seconds, 60℃ for 30 seconds, and then 72℃ for 30 seconds were followed with an extension for 5 minutes at 72℃ for 5 minutes and held the reaction at 10℃

The PCR Master Mix and PCR primer Cocktail tubes were centrifuged at 600×g for 5 seconds.

The PCR plate was removed from -20℃ and incubated at room temperature. Prior to use, centrifuged the thawed plate at 280×g for 1 minute. The adhesive seal was removed.

The 96-well PCR plate was labeled as target sample plate (TSP1 plate).

1.To make the PCR plate

5 µl of thawed PCR Primer Cocktail was added to each well of PCR plate. Then, 25µl of thawed PCR Master Mix was added. The mixture was gently pipetted up and down ten times to mix it thoroughly. Then, the PCR plate was sealed as above.

2.Amplification of PCR plate

The sealed PCR plate was placed on a preheated thermocycler and the program named PCR was run. Then, the plate was removed from the thermocycler when it reached 10°C. After removal, the adhesive seal was removed from the PCR plate.

3.Cleaning up the PCR plate:

50 µl well-dispersed AMPure XP beads were added to each well of the PCR plate. The mix was gently pipetted up and down ten times to mix it thoroughly. The mix was then incubated at room temperature for 15 minutes. The PCR plate was then placed on the magnetic stand for five minutes at room temperature. Then 95 µl supernatant was removed from each well of the plate and then discarded. While on the magnetic stand, each well of the PCR plate was filled with 200 µl of freshly prepared 80% ethanol without touching the beads. The PCR plate was then again incubated at room temperature for 30 seconds and then all supernatants were discarded. This was repeated once again for a total of two ethanol washes. After the wash, the plate was incubated at room temperature for 15 minutes to dry and then it was removed from the magnetic stand. Prior to use, the resuspension buffer was centrifuged for five seconds at 600×g and then 32.5 µl of it was added to each well of the PCR plate. The mix was gently pipetted up and down ten times to mix it thoroughly. Then incubation was done at room temperature for two minutes. After incubation, the plate was placed on the magnetic stand for five minutes at room temperature. Then, 30 µl of the supernatant from each well of the PCR plate was transferred to the corresponding well of the new 0.3 ml PCR plate labeled as TSP1 plate.

VIII) Validation of library

It is suggested by Illumina, Inc that for validation, quality and quantity should be controlled by using Agilent technologies 2100 Bioanalyzer (For example: Agilent DNA-1000) and qPCR by following Illumina Sequencing Library qPCR Quantification Guide(Illumina, Inc) respectively. However, there are alternative methods that can be followed as well.

For quality control, samples can be run on a 1% Agarose gel.

For quantity, Qubit fluorometer was used which uses fluorescence dye to determine the concentration of nucleic acid. Qubit dsDNA high-sensitivity (HS) assay was followed for the quantification of the cDNA library.

Calibration performance using Qubit dsDNA high-sensitivity (HS) assay (Thermofisher Scientific, USA):

Qubit working solution was made by diluting Qubit® dsDNA HS Reagent 1:200 in Qubit® dsDNA HS Buffer.in a clean 0.5mL PCR tubes. For HS assay we need two standard solutions. For that, two tubes each one containing 190 µl of Qubit working solution were prepared. Then, 10 µl of each Qubit standard to individual tube was added, and labeled as Standard 1, & Standard 2. For samples, in 6 tubes, 199 µl of Qubit working solution was added and 1 µl of each sample. After vortexing for 2-3 seconds, all tubes were incubated for two minutes at room temperature.

Reading standards and samples at “Qubit® 3.0 Fluorometer”

On the home screen of the fluorometer, DNA button was pressed and dsDNA high sensitivity assay was selected afterwards.

Then, when read standards displayed, standard 1 tube was inserted in the sample chamber and closed the lid and then pressed the read standard. When it showed reading is complete after 3 seconds, standard 1 tube was removed and then inserted the 2nd standard and followed the same process. Once the reading of standards was completed, the screen will show: Run samples. After pressing, we have an option to select the sample volume, which was set up as 1 µl. after that the read sample button was pressed. The screen displayed the result: top value showed the original concentration in the sample and the bottom one showed the diluted concentration.

https://tools.thermofisher.com/content/sfs/manuals/Qubit_dsDNA_HS_Assay_UG.pdf

IX)Normalization and pooling of libraries

Prior to normalizing the libraries, it was necessary to convert the concentration of samples in each well from ng/µl to nM. For that, the following two things were determined: the average size of all the samples in the library and the concentration from the Qubit (Fisher Scientific, USA). Once determined the following equation was used to convert the concentration to nM:

$$\frac{(\text{Concenration in ng/µl})}{(660\,\text{g/mol} \times \text{average library size in bp})} \times 10^6 = \text{concentration in nM}$$

https://support.Illumina.com/bulletins/2016/11/converting-ngl-to-nm-when-calculating-dsdna-library-concentration-.html

1.Normalization of libraries

In a library, each sample in the well has a different concentration. So, it is important to normalize the concentration for all samples that need to be sequenced using Miseq.

To make a library concentration 4nM this simple formula has been used:

C1V1=C2V2

Where,

C1 is the initial concentration of the sample

V1 is the required volume

C2 is the final concentration of the sample

V2 is the final volume that required

2.Pooling of libraries

Once all samples are normalized to 4 nM, the next step is pooling of all samples in a single tube.10 µl of each sample (4 samples: C2, C3 and S1, S3) of the library was collected and pooled into one microfuge tube. The total volume was 40 µl which was vortexed for five seconds so that samples were mixed well. Then 5 µl of the mixture was taken to a new tube and then added 5 µl of 0.2M NaOH. (The M1V1=M2V2 formula was followed to make 0.2 M NaOH from 1M NaOH) Then again, vortexed the mix for five seconds then incubated it for 5 minutes at room temperature. 990 µl of HT1 (from Truseq Kit) was added. Finally, this gave 1ml of 20pM of library. To make the library of 8pM concentration, 240 µl of the sample from the 20 pM library sample was taken and added 360µl of HT1 solution. 600µl of the total sample was now ready to transfer to the cartridge for sequencing.

MiSeq Reagent Kit v3 was used for which usually gives sequencing with increased cluster density and read length with a good quality (Q) score (Illumina, Inc).

Chapter 3

Results and Discussion

Physiological studies

Under drought stress, plants change some mechanisms to adapt to the shortage of water as well as nutrients (Ben-Jabeur et al. 2021). Plants under drought stress reduce linear electron flow (LEF) (Ben-Jabeur et al. 2021) to protect PSII and PSI (Huang et al. 2012). In the linear electron flow of oxygenic photosynthesis, electrons from the oxygen-evolving complex are transferred to $NADP^+$, reducing it to NADPH (Huang et al. 2018). There are several mechanisms to protect PSII: activation of non-photochemical quenching (NPQ), activation of antioxidants to scavenge ROS species, development of cyclic electron flow, alternative electron flow, and photorespiration (Bashir et al. 2021 & Muhammad et al. 2021). Reactive Oxygen Species can be produced as the result of over-reduction of reaction centers due to excess light, which is not adequately dissipated (Ben-Jabeur et al. 2021 & Loggini et al. 1999). In our studies, the analysis of variance (ANOVA) suggested that drought stress has a significant effect on some traits of plants, such as NPQ, Phi2, PhiNO, PhiNPQ, ECSt, Fv´/Fm´, Leaf temperature, and LEF (Table 6; Table 7). The water shortage increased the PhiNPQ and NPQ while decreasing Phi2, PhiNO, PhiNPQ, ECSt, Fv´/Fm´, and LEF (Figure 10).

Phi2 parameter measures captured light energy by photosystem II which is used for ATP and NADPH production in the sugar-phosphate mechanism for plant growth. PhiNPQ is the portion of the light that is not used in PSII and scattered towards the non-photochemical quenching as heat. The photo energy lost in the photosystem process can

be termed PhiNO. Linear electron flow (LEF)can be calculated from Phi2(PhotosynQ.com/tutorial).

The decrease in Electrochromic Shift (ECSt gives an idea about ATP synthase capacity) is mainly because of the reduction of intercellular carbon dioxide concentration due to stomatal closure and chlorophyll degradation under drought stress which leads to lower production of ATP and NADPH, which reduces the linear electron flow (LEF) (Ben-Jabeur et al. 2021). The limitation of CO_2 can lead to a decrease in PSII redox site and its operating efficiency. This could be a reason for the reduction of quantum yield of PSII (Phi2), Fv′/Fm′ (The maximal quantum efficiency of PSII) (Ben-Jabeur et al. 2021 & PhotosynQ.com/tutorial). Phi2 under water limitation is associated with increased PhiNPQ and NPQ and decreased PhiNO to scavenge ROS, reduce damage, and adapt the plant under prolonged abiotic stress (Ben-Jabeur et al. 2021).

Drought stress imposes a degradation of chlorophyll a, chlorophyll b, and total chlorophyll in plants which can be considered one crucial trait of oxidative stress and degradation of chlorophyll (Anjum et al. 2011). The reports in the work presented here showed a significant reduction of chlorophyll a, chlorophyll b, and total chlorophyll (Figure 12; Figure 13), which is similar to the report given by Mafakheri et al. 2010 who reported the decrease in chlorophyll a, chlorophyll b, and total chlorophyll in three chickpea cultivars. Chlorophyll, one of the essential contents of chloroplast, has a direct relationship with the photosynthesis ability of plants (Anjum et al. 2011). The lowered chlorophyll content in drought-stressed plants can cause the degradation of the photosynthetic rate (Figure 14). The degradation of chlorophyll content can also be due to

leaf senescence which ultimately leads to lipid peroxidation in the chloroplast (Dhindsa et al. 1981).

Relative moisture content (RMC) in drought-stressed plants is a tolerance indicator (Maghsoudi et al. 2016). Higher moisture content in drought stress has been reported in wheat cultivars, indicating stress tolerance of wheat against drought stress (Maghsoudi et al. 2016). The difference in relative moisture content can be due to two reasons: either the ability of the plant to uptake water from soil or stomatal conductance in response to the shortage of water (Keyvan 2010). In this study, RMC was decreased under drought stress (Figure 16). However, when testing the data by ANOVA, the difference was not significant, indicating turmeric plants did not show any effect on moisture content under drought stress (Figure 15).

The catalase activities determine the abundance of O_2, H_2O_2, OH^- & O_2^- (Dhindsa et al. 1981). These free radicals (O_2, H_2O_2, OH^- & O_2^-) are proven to have an ability to initiate lipid peroxidation in senescence leaves (Fong et al. 1973). Lipid peroxidation means cell membrane degradation under harsh environments. The lipid peroxidation lowers catalase activity and indicates the senescence of the stressed plant cells (Dhindsa et al. 1981). In this study, drought-stressed plants showed a decline in catalase activity (Figure 17). Since there is not enough catalase activity to scavenge the free radicals, it is assumed that the free radicals initiated lipid peroxidation in the turmeric leaves. Therefore, we assumed that the turmeric plants may reach senescence within 21 days of the stress.

Table 6: ANOVA of photosynthetic traits between the controlled and drought-stressed plant groups

ANOVA	Traits				
	ECSt	Fv′/Fm′	Leaf Temperature	Leaf Thickness	LEF
F value	6.839**	525.1	5.09	0	23.375
P-value	0.018*	1.163e-13 ***	0.383*	0.996	0.0001***

Table 7: ANOVA of photosynthetic traits between the controlled and drought stressed plant groups

ANOVA	Traits				
	NPQt	Phi2	PhiNO	PhiNPQ	qL
F value	246.21	362.93	368.25	426.77	1.9744
P-value	3.882e-11 ***	2.023e-12 ***	1.809e-12 ***	5.804e-13 ***	0.1791

One way ANOVA analysis of 3 controlled plants and 3 drought stressed plants of different parameters collected using MultispeQ: - total non-photochemical quenching (NPQt) , quantum yield of PSII (Phi2), fraction of light lost via nonregulated photosynthesis inhibitor processes (PhiNO), fraction of light dedicated to non-photochemical quenching (PhiNPQ), fraction of PSII open centers (qL), total electrochromic shift (ECSt), maximal quantum efficiency of PSII (Fv′/Fm′), Leaf temperature, Leaf thickness, linear electron flow (LEF).

*Symbols indicate the statistical significance: *=$p \leq 0.05$, ** =$p \leq 0.01$ & ***=$p \leq 0.001$

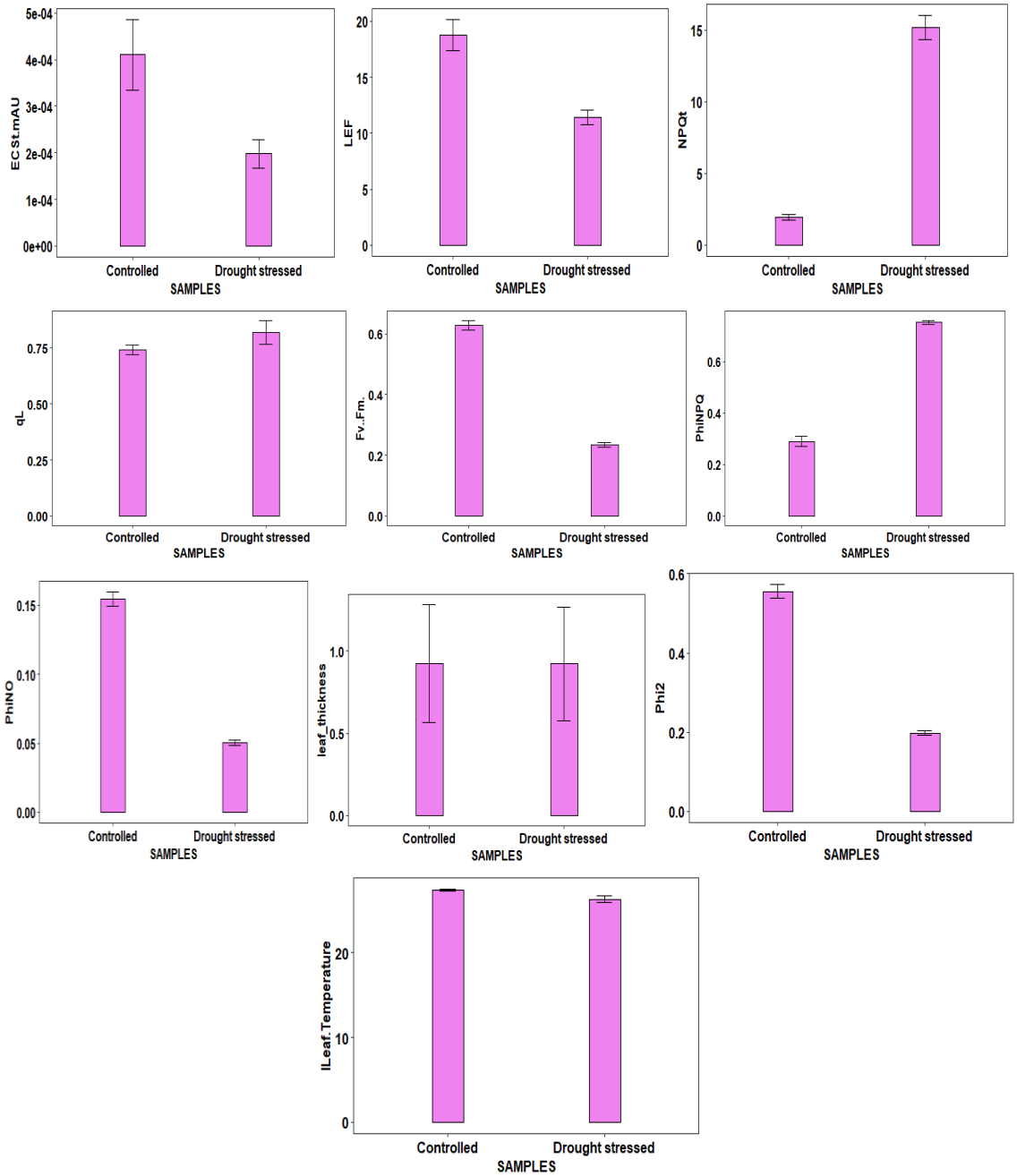


Figure 10: This figure shows the data, which are the means calculated from three technological replicates from each biological replicates. - total non-photochemical quenching (NPQt) , quantum yield of PSII (Phi2), fraction of light lost via nonregulated photosynthesis inhibitor processes (PhiNO), fraction of light dedicated to non-photochemical quenching (PhiNPQ), fraction of PSII open centers (qL), total electrochromic shift (ECSt), maximal quantum efficiency of PSII (Fv′/Fm′), Leaf temperature, Leaf thickness, linear electron flow (LEF).

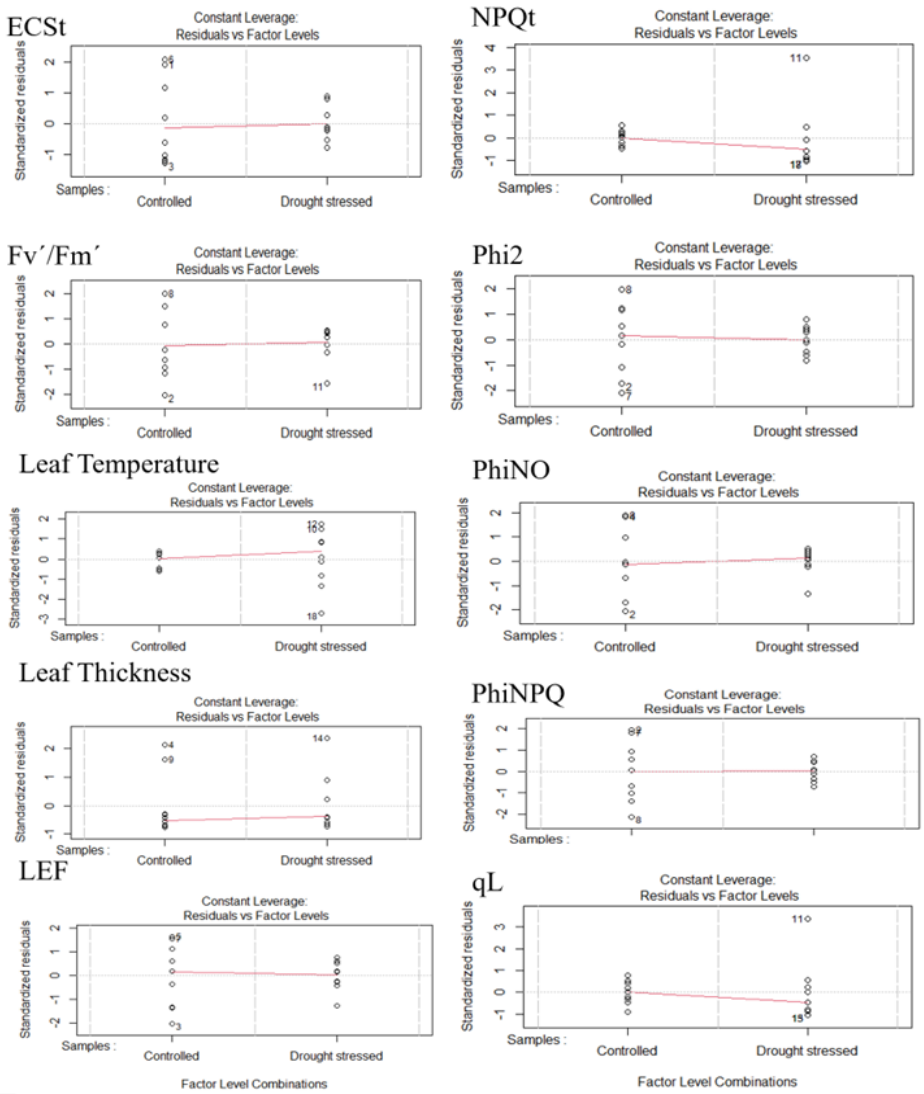


Figure 11: This figure shows the Residuals vs. Fitted model from the data calculated from three technological replicates from each biological replicates. - total non-photochemical quenching (NPQt) , quantum yield of PSII (Phi2), fraction of light lost via nonregulated photosynthesis inhibitor processes (PhiNO), fraction of light dedicated to non-photochemical quenching (PhiNPQ), fraction of PSII open centers (qL), total electrochromic shift (ECSt), maximal quantum efficiency of PSII (Fv′/Fm′), Leaf temperature, Leaf thickness, linear electron flow (LEF).

```
> anova(model2)
Analysis of Variance Table

Response: Total.chlorophyll
          Df Sum Sq Mean Sq F value   Pr(>F)
Samples    1   3209  3209.0  9.2392 0.004022 **
Residuals 43  14935   347.3
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
```

Figure 12: ANOVA of Total Chlorophyll between the controlled and drought stressed plant groups. *Symbols indicate the statistical significance: *=p≤0.05, ** =p≤0.01 & ***=p≤0.001

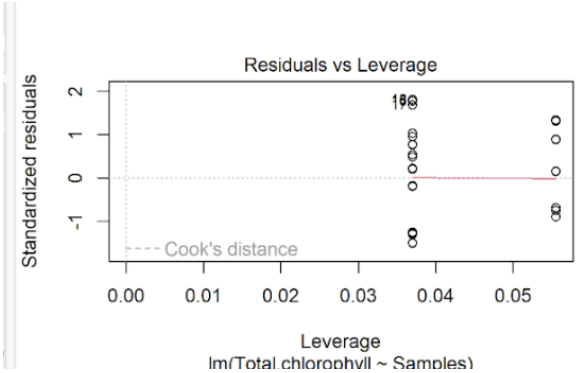


Figure 13: Residuals vs. Fitted model to detect linearity, variances, and outliers of Total Chlorophyll Content between the controlled and drought stressed plant groups

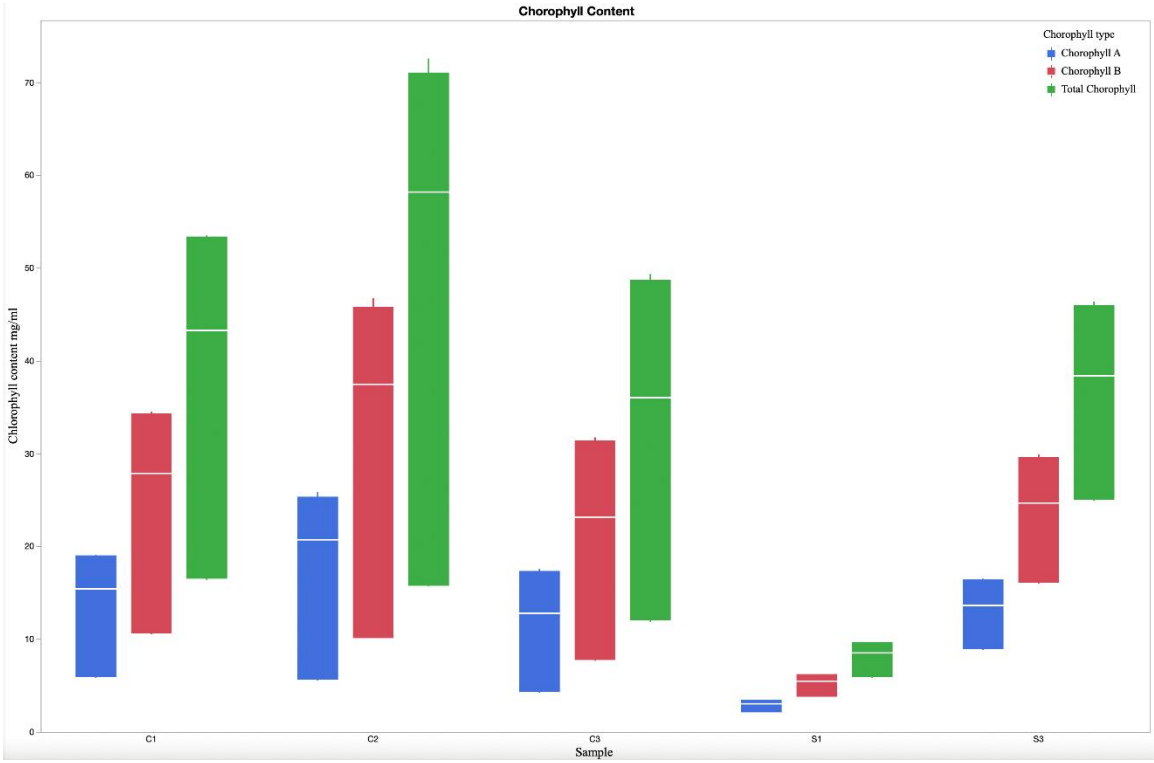


Figure 14: This graph depicts 3 technical replicates in each biological replicate of the total chlorophyll content present. Control: n = 3, and Drought: n = 2. The bars above the graphical bars represent standard error, and white line in each bar shows the mean of three replicates

```
> anova(model2)
Analysis of Variance Table

Response: Relative.Moisture.Content
          Df  Sum Sq Mean Sq F value Pr(>F)
Samples    1 0.11188 0.11188  3.0092 0.1812
Residuals  3 0.11154 0.03718
```

Figure 15: ANOVA of Total Chlorophyll between the controlled and drought stressed plant groups

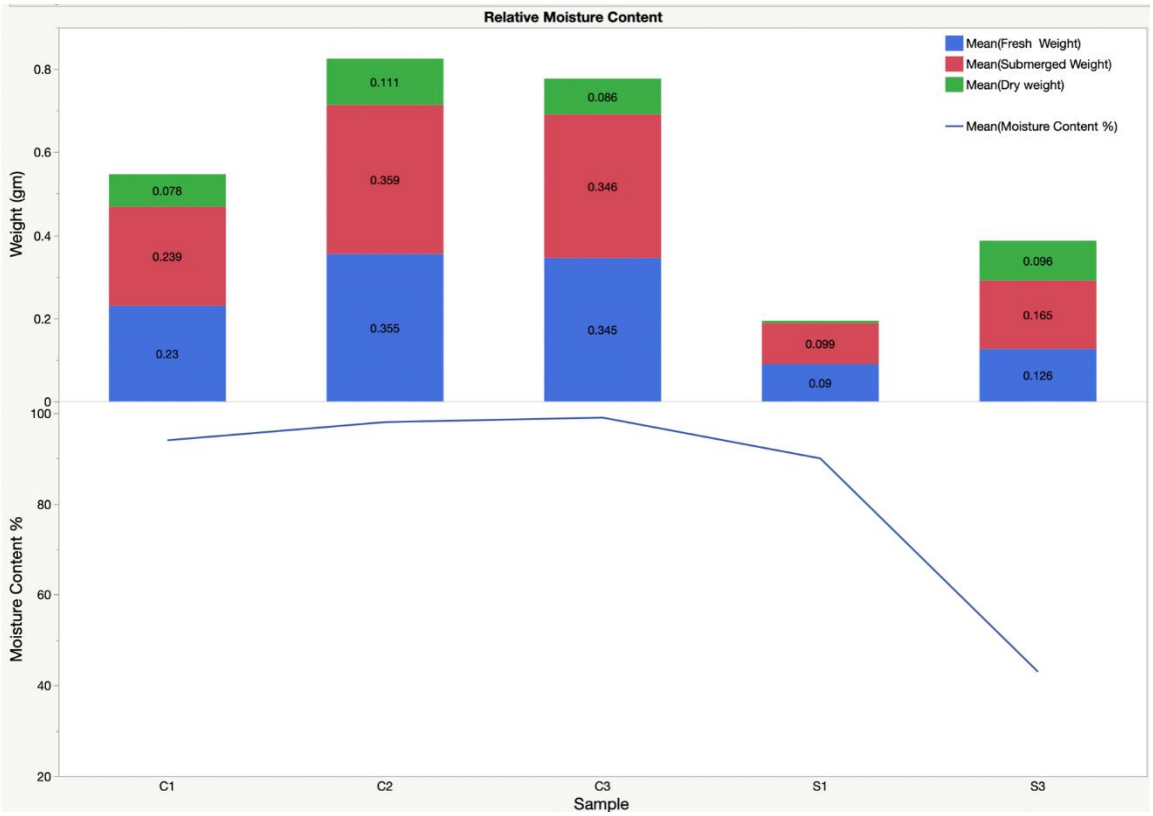


Figure 16: This first figure displays the mean of fresh weight, submerged weight and dry weight of leaves of each biological replicate. The below figure depicts relative moisture content (%) in the leaves of each biological replicate

Total number of biological replicates: under drought conditions (n = 2), and under control conditions (n = 3).

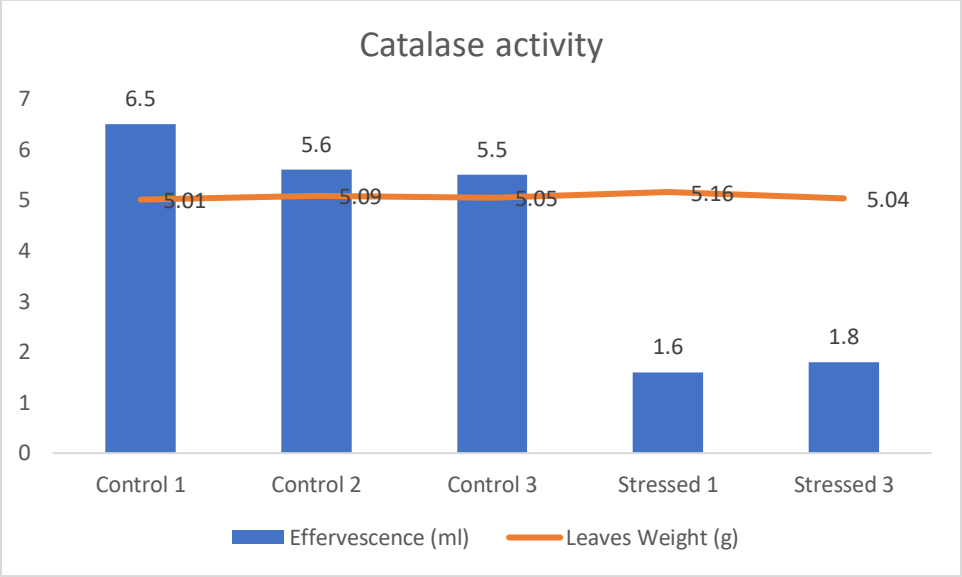


Figure 17: This graph depicts the Catalase activity of biological replicates. Control: n = 3, and Drought: n = 2

Molecular studies

RNA isolation is essential in molecular biology as it is responsible for further gene expression studies and transcriptomic analysis. In this study, RNA was isolated to perform RNAseq-transcriptomic analysis (Table 8). So, it was important to get pure RNA. RNA isolation from turmeric is difficult as it has a lot of secondary metabolites that bind to the single strand or are co-precipitated with RNA (Ghawana et al, 2011). The Trizol method given by Ramadoss and Basu, 2018 was used instead of commercial kits. Also, for purification, RNA clean and concentrator kits were used. The RNA quality was checked using RNA formaldehyde-free gel electrophoresis (VWR Life Science, USA) (Figure 21). The quantity was assessed at $A_{260/280}$ ratio and $A_{260/230}$ ratio in a NanoDrop UV Visible Spectrophotometer (Thermo Scientific, USA) (Figure 20). In this study, several RNA extractions gave low $A_{260/230}$ ratio indicating the presence of a high amount of polyphenols in turmeric (Figure 18 & Figure 19) (https://medicine.yale.edu/).

Table 8: RNA samples that were used for further analysis

RNA samples	Concentration(ng/µl)	$A_{260/280}$ratio
C1-3	67.1	2.15
C2-3	101.90	1.84
C3-3	350.50	2.10
S1-2	28.2	2.04
S2-1	21.8	1.72
S3-3	72.2	2.16

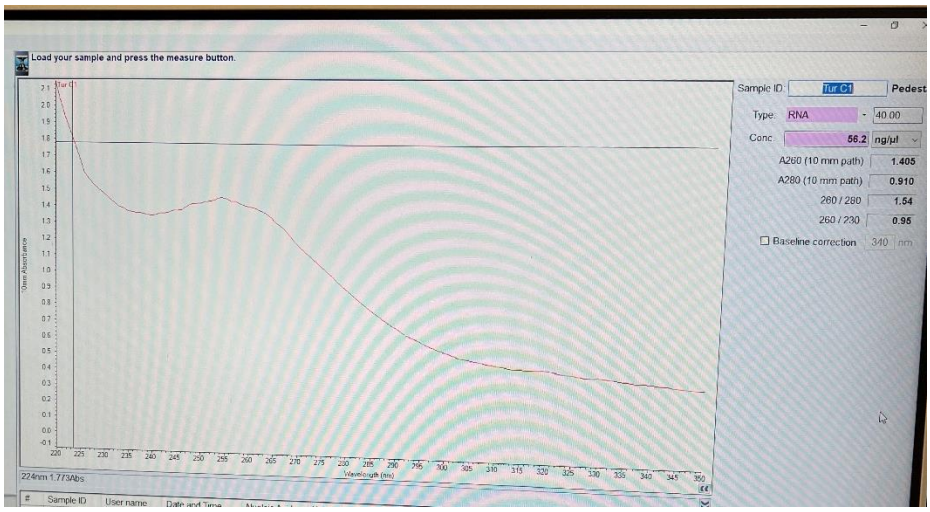
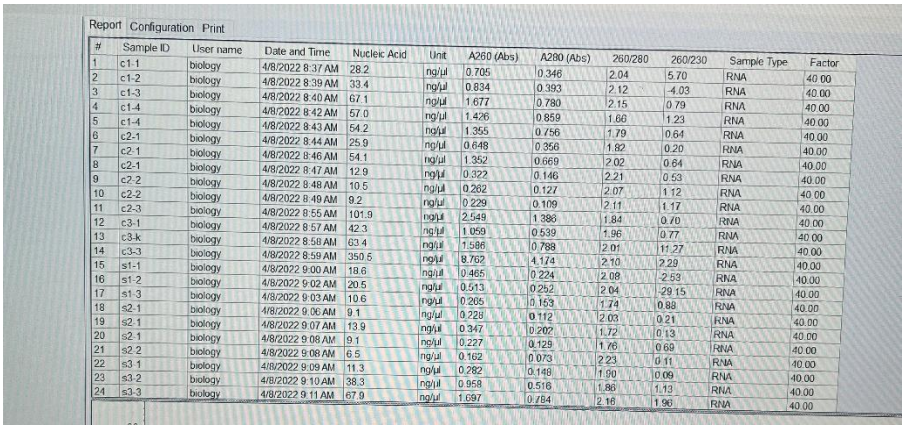


Figure 18: RNA isolation from turmeric with a low $A_{260/230}$ ratio



Report Configuration Print

#	Sample ID	User name	Date and Time	Nucleic Acid	Unit	A260 (Abs)	A280 (Abs)	260/280	260/230	Sample Type	Factor
1	c1-1	biology	4/8/2022 8:37 AM	28.2	ng/µl	0.705	0.346	2.04	5.70	RNA	40.00
2	c1-2	biology	4/8/2022 8:39 AM	33.4	ng/µl	0.834	0.393	2.12	-4.03	RNA	40.00
3	c1-3	biology	4/8/2022 8:40 AM	67.1	ng/µl	1.677	0.780	2.15	0.79	RNA	40.00
4	c1-4	biology	4/8/2022 8:42 AM	57.0	ng/µl	1.426	0.859	1.66	1.23	RNA	40.00
5	c1-4	biology	4/8/2022 8:43 AM	54.2	ng/µl	1.355	0.756	1.79	0.64	RNA	40.00
6	c2-1	biology	4/8/2022 8:44 AM	25.9	ng/µl	0.648	0.356	1.82	0.20	RNA	40.00
7	c2-1	biology	4/8/2022 8:46 AM	54.1	ng/µl	1.352	0.669	2.02	0.64	RNA	40.00
8	c2-1	biology	4/8/2022 8:47 AM	12.9	ng/µl	0.322	0.146	2.21	0.53	RNA	40.00
9	c2-2	biology	4/8/2022 8:48 AM	10.5	ng/µl	0.262	0.127	2.07	1.12	RNA	40.00
10	c2-2	biology	4/8/2022 8:49 AM	9.2	ng/µl	0.229	0.109	2.11	1.17	RNA	40.00
11	c2-3	biology	4/8/2022 8:55 AM	101.9	ng/µl	2.549	1.386	1.84	0.70	RNA	40.00
12	c3-1	biology	4/8/2022 8:57 AM	42.3	ng/µl	1.059	0.539	1.96	0.77	RNA	40.00
13	c3-k	biology	4/8/2022 8:58 AM	63.4	ng/µl	1.586	0.788	2.01	11.27	RNA	40.00
14	c3-3	biology	4/8/2022 8:59 AM	350.5	ng/µl	8.762	4.174	2.10	2.29	RNA	40.00
15	s1-1	biology	4/8/2022 9:00 AM	18.6	ng/µl	0.465	0.224	2.08	2.53	RNA	40.00
16	s1-2	biology	4/8/2022 9:02 AM	20.5	ng/µl	0.513	0.252	2.04	29.15	RNA	40.00
17	s1-3	biology	4/8/2022 9:03 AM	10.6	ng/µl	0.265	0.153	1.74	0.88	RNA	40.00
18	s2-1	biology	4/8/2022 9:06 AM	9.1	ng/µl	0.228	0.112	2.03	0.21	RNA	40.00
19	s2-1	biology	4/8/2022 9:07 AM	13.9	ng/µl	0.347	0.202	1.72	0.13	RNA	40.00
20	s2-1	biology	4/8/2022 9:08 AM	9.1	ng/µl	0.227	0.129	1.76	0.69	RNA	40.00
21	s2-2	biology	4/8/2022 9:08 AM	6.5	ng/µl	0.162	0.073	2.23	0.11	RNA	40.00
22	s3-1	biology	4/8/2022 9:09 AM	11.3	ng/µl	0.282	0.148	1.90	0.09	RNA	40.00
23	s3-2	biology	4/8/2022 9:10 AM	38.3	ng/µl	0.958	0.516	1.86	1.13	RNA	40.00
24	s3-3	biology	4/8/2022 9:11 AM	67.9	ng/µl	1.697	0.784	2.16	1.96	RNA	40.00

Figure 19: RNA extractions each biological replicates: six biological replicates (three controlled groups and three drought-stressed groups)

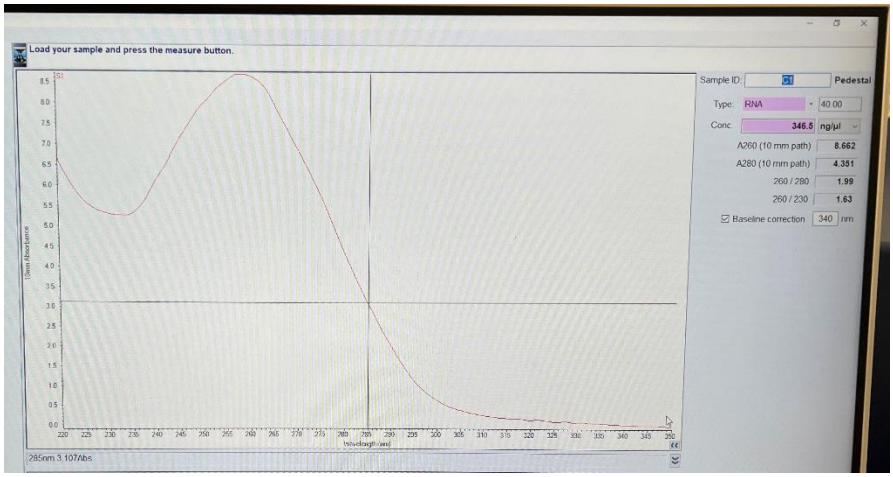


Figure 20: RNA isolation with a high concentration and expected $A_{260/280}$ ratio and $A_{260/230}$ ratio in Nanodrop was used for further downstream process.

Agarose formaldehyde-free gel was used to assess the quality of the RNA (Figure 21).

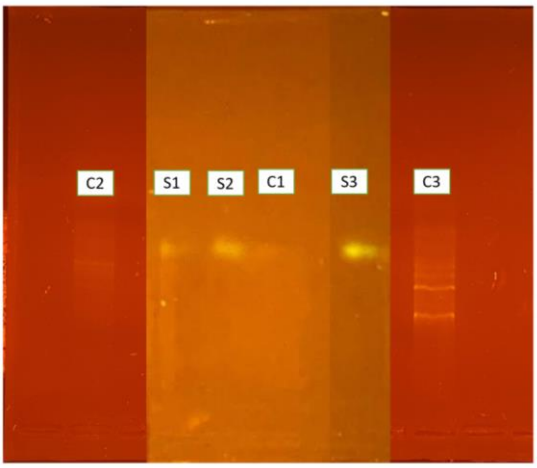


Figure 21: RNA Agarose formaldehyde free gel of 3 controlled samples and 3 drought stressed samples of turmeric plants. Gels were run on separate days.

After the cDNA library preparation, a Qubit® 3.0 Fluorometer was used to quantify the data (Table 9) and the concentration of the libraries were determined (Table 10)

Table 9: Qubit® 3.0 Fluorometer quantification

Samples	Concentration (ng/µl)	Dilution factor
C1	0.16	200
C2	0.732	200
C3	0.682	200
S1	1.01	200

S2	0.196	200
S3	2.74	200

Table 10: The concentration of cDNA library in nM

Sample Id	Concentration (nM)
C1	1.16
C2	2.91
C3	2.33
S1	4.14
S2	1.17
S3	5.60

Since, the concentration of the library was not high enough, everything was normalized to 2nM. Therefore, C1 &S2 were eliminated from the pool due to low concentration (Table 11).

Table 11: Normalization of 4 samples of cDNA using C1V1=C2V2 formula

Samples	Final concentration (C2) (nM)	Final volume (V2) (µL)	Initial concentration (C1) (nM)	Initial Volume(v1) (µL)
C2	2	15	2.91	10.30
C3	2	15	2.33	12.87
S1	2	15	4.14	7.24
S3	2	15	5.60	5.36

Sequencing analysis of raw data from BaseSpace

Sequencing By Synthesis (SBS) data from MiSeq instruments gets automatically uploaded to a website called BaseSpace. The SBS data then gets converted to .bcl (basecell) files. Then, according to adapter indexes, samples get separated, which is called demultiplexing. This data is always received in FastQ format.

For this project, total reads and individual read yields placed in the Basespace hub. The run showed a pass with a Q30 average score of 86.93% and (Figure 22 & Table 12). The Q30 score reflects at confidence level of 99.99% of each read.

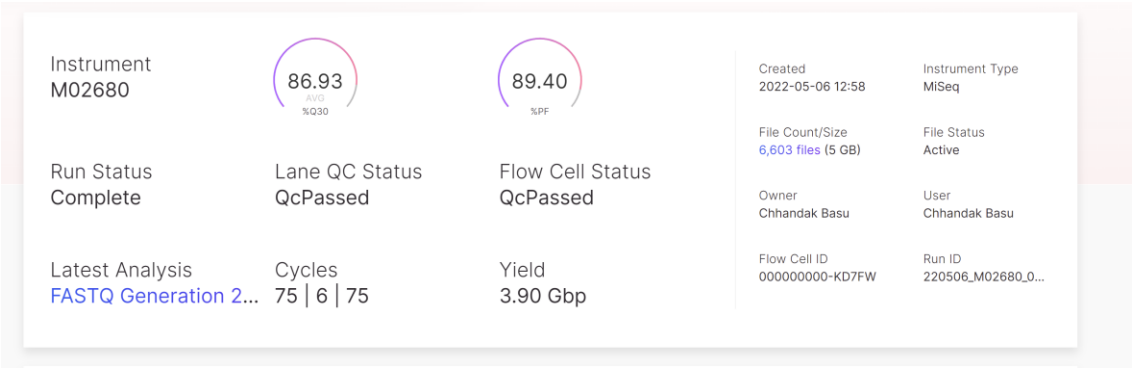


Figure 22: A screenshot taken directly from the Basespace sequencing hub, showing the run status, total yield and Q30 score

Table 12: Samples with read yield for each individual species

Samples	Yield
C2	940.24 Mbp
C3	1.10 Gbp
S1	958.20 Mbp
S3	602.13 Mbp

The quality control was performed by using an app from BaseSpace Labs, which is called FastQC. FastQC reports the collective quality of all reads rather than individual read quality. This step is important to run in order to get some insights into the raw data that comes from high sequence throughput. With the help of this analysis, any problems in the data can be seen before going for further analysis. FastQC reports per base sequence quality, per base quality scores, per base sequence content, per base GC content, per sequence GC content, per base N content, sequence length distribution, sequence duplication levels and overrepresented sequences.

(https://basespace.Illumina.com/apps/706706/FastQC)

per base sequence quality: This image shows the overview of the range of quality of all bases at each position. In this figure (Figure 23), some sudden drops in quality ranges can be seen and aremostly due to instrumental breakdown. (https://hbctraining.github.io/Intro-to-rnaseq-hpc salmon/lessons/qc_fastqc_assessment.html)

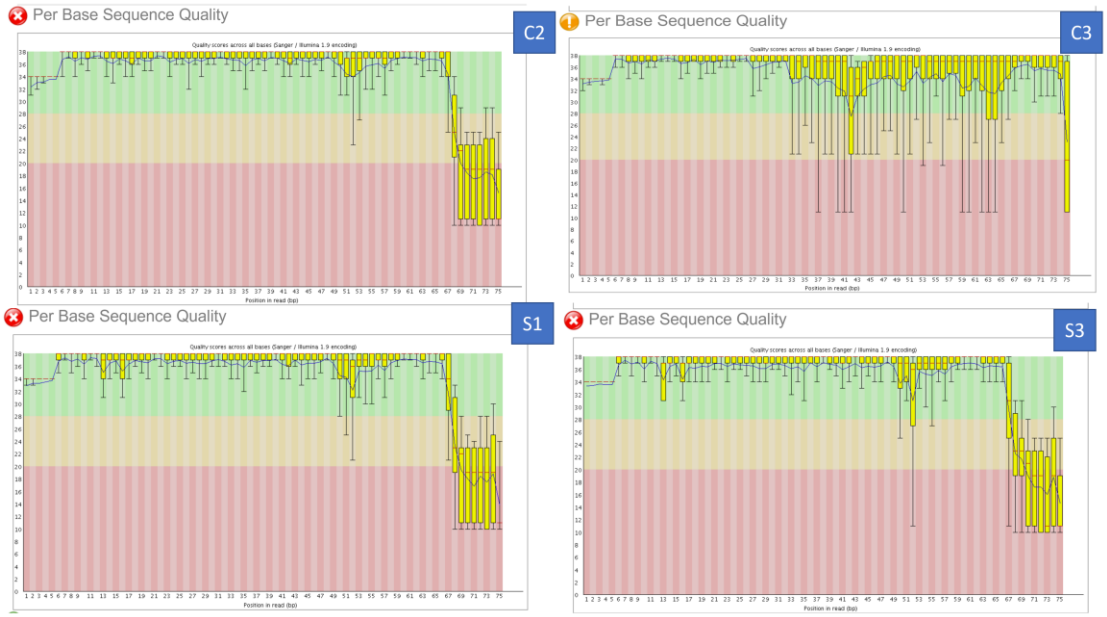


Figure 23: This is showing the per base sequence quality of reads of two controlled turmeric sample (C2 &C3) and two drought stressed sample (S1 &S3) Screenshots were obtained from http://www.illiumina.basespace.com.

per sequence quality score: This gives the average quality score on X-axis, and the Y-axis gives the sequences with the average quality (Figure 24).

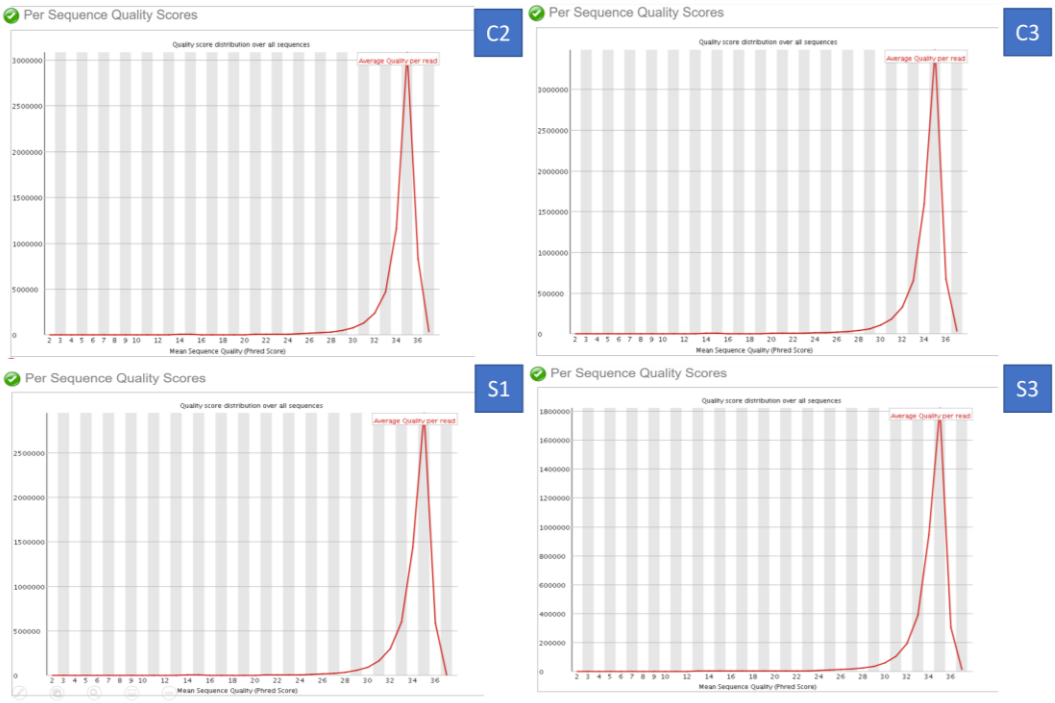


Figure 24: This is showing the per sequence quality scores of reads of two controlled turmeric sample (C2 &C3) and two drought stressed sample (S1 &S3) Screenshots were obtained from http://www.illiumina.basespace.com.

per base sequence content: This gives the proportion of each base position to four nucleotides in a file. Normally, there will be no difference between the four DNA bases in sequences, and the four lines should be parallel to each other. (https://www.bioinformatics.babraham.ac.uk/projects/fastqc/Help/3%20Analysis%20Modules/4%20Per%20Base%20Sequence%20Content.html)

However, in our RNAseq library there is a fail due to the presence of random hexamers in the library (Figure 25). (https://hbctraining.github.io/Intro-to-rnaseq-hpc-salmon/lessons/qc_fastqc_assessment.html)

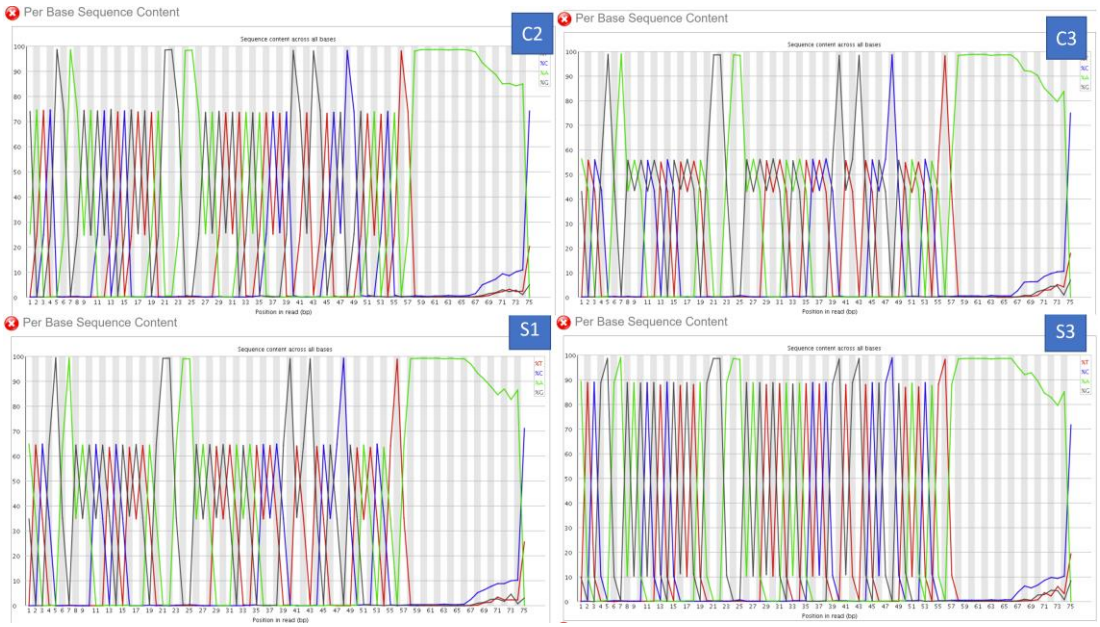


Figure 25: This is showing the per base sequence content of reads of two controlled turmeric sample (C2 &C3) and two drought stressed sample (S1 &S3) Screenshots were obtained from http://www.illiumina.basespace.com

per sequence GC content: This plot gives the GC content of the whole sequence of a sample and then compares it to the model GC content in a normal distribution. If we see more than one peak in the center, it can be due to contamination with other species. In this plot, only one peak in all four samples was seen, which represented the absence of contamination, but those high peaks are due to overrepresented sequences. (Figure 26).

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/Help/3%20Analysis%20Modules/5%20Per%20Sequence%20GC%20Content.html

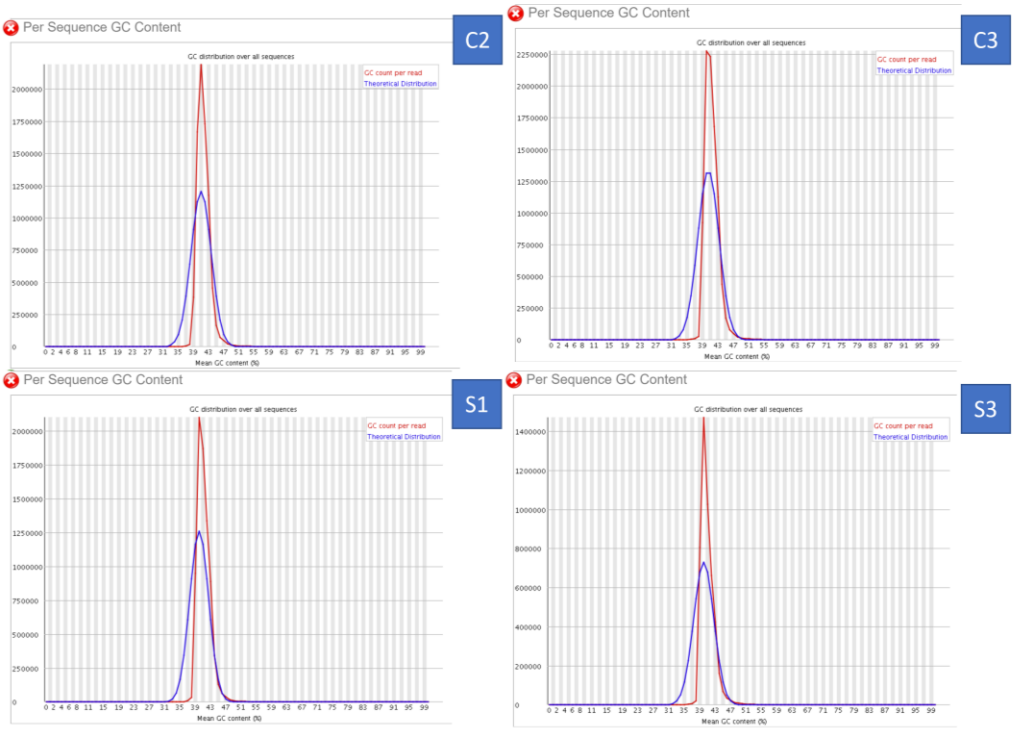


Figure 26: This is showing the per sequence GC content of reads of two controlled turmeric sample (C2 &C3) and two drought stressed sample (S1 &S3) Screenshots were obtained from http://www.illiumina.basespace.com

So for this RNAseq data, the FastQC reported failed per base sequence quality, per base quality scores, per base sequence content, per base GC content, per sequence GC content, per base N content.

After trimming of adapters, the reads were not enough to de novo analysis (Table 13 & Figure 27). To go for a de novo assembly would require enormous computational time to produce a usable reference genome/transcriptome for further analysis. If we could make 50M reads per sample, it may be possible to do a de-novo assembly.

Table 13: Read yield for each individual sample after trimming of adapters

Samples	Read yield after trimming
C2	0.069M
C3	0.068M
S1	0.025M
S3	0.035M

```
TC2.trim_1P.fastq.gz
.06912100000000000000
TC2.trim_2P.fastq.gz
.06912100000000000000
TC3.trim_1P.fastq.gz
.06875700000000000000
TC3.trim_2P.fastq.gz
.06875700000000000000
TS1.trim_1P.fastq.gz
.02502100000000000000
TS1.trim_2P.fastq.gz
.02502100000000000000
TS3.trim_1P.fastq.gz
.03547900000000000000
TS3.trim_2P.fastq.gz
.03547900000000000000
```

Figure 27: This shows total amount of reads after trimming the adapters of two controlled turmeric sample (C2 &C3) and two drought stressed sample (S1 &S3)

Some papers where researchers did perform genomic sequencing on turmeric and transcriptome have been studied extensively. Unfortunately, the authors of the paper (Chakraborty et al. 2021) did not deposit any assembled reference genome and transcriptome into the public domain database. In the future, if someone deposits any reference genome for *Curcuma longa* since more research is being done on turmeric, the produced data from this project can be used to do a transcriptomic analysis of turmeric under drought stress. Also, as mentioned above, ginger has a close relationship with turmeric (Figure 7). So, in the future, it is also considered to use the zinger reference genome to analyze the available data.

Future studies of turmeric: the de-novo assembly of non-model species

Turmeric is a non-model organism. Analyzing RNA-seq data for non-model species is challenging due to the unavailability of the reference genome and the absence of tools for

further data analysis (Chalifa-Caspi 2021). For the RNAseq analysis, we need a strategic plan to search the available draft genome for the specific organism, clean the raw reads for de-novo assembly, carry out the de novo assembly, and assess transcriptomic analysis by quantifying transcriptomes (Chalifa-Caspi 2021). For now, a user-friendly workflow with all tools used for non-model species by non-programmers is crucial.

De novo genome sequence assembly of Next Generation Sequencing (NGS) is essentially the reconstruction of a complete genome from short sequence reads without the help of a reference genome. It assembles raw reads into contigs and then contigs to scaffolds (Liao et al. 2019) (Figure 28).

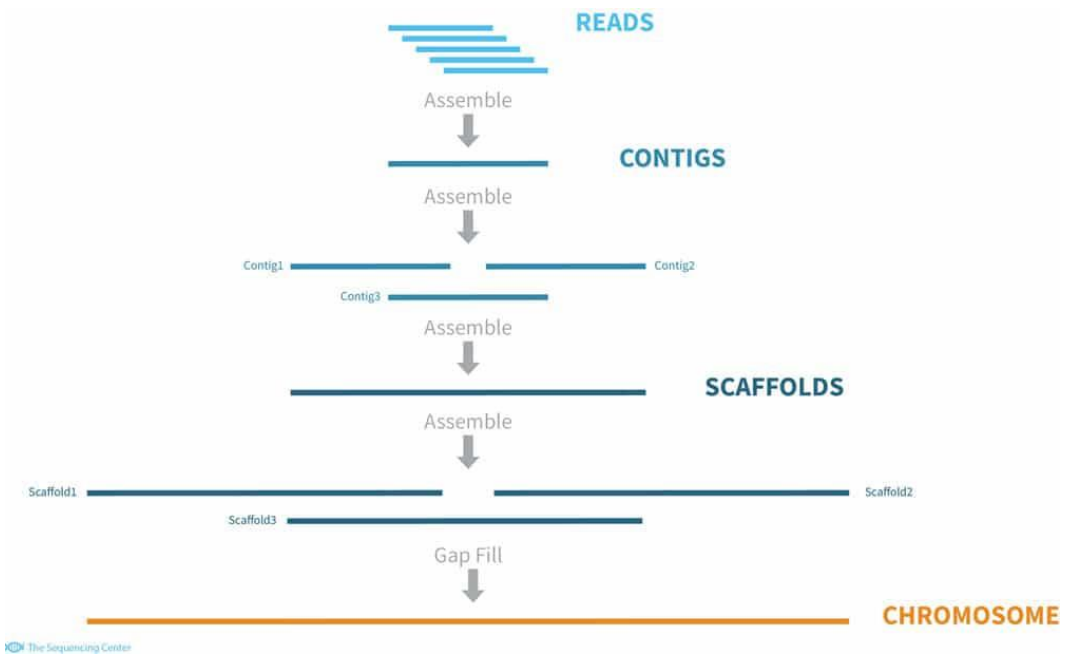


Figure 28: A step by step process of de novo sequence assembly from raw data of Next Gen Sequencing (NGS) Source: https://thesequencingcenter.com/knowledge-base/de-novo-assembly/

There are some challenges while performing de-novo assembly. One of the challenges is sequencing error (Liao et al. 2019). It mostly happens towards the end of the reads due to substitutions or deletion errors (Heydari et al. 2019). Trimming could be a solution for this type of error, but trimming reduces read length (Heydari et al. 2019). Another

challenge is computational cost with a long time with high RAM storage. For unusually long genomes, they take several days to weeks with colossal RAM storage (Liao et al. 2019). Due to shorter reads produced by Illumina Miseq sequencing (Second generation sequencing technology) and sequencing error, the de novo assemble process is challenging for non -programmers.

Hypothesized gene and their pathways that get affected by drought stress

There have been some studies of miRNA of turmeric under drought stress. These evolutionary conserved non-coding RNA molecules are responsible for targeting mRNA and binding to them to prevent protein production by one of two distinct mechanisms (Macfarlane & Murphy et al. 2010). One such miRNA that has been differentially expressed in turmeric under drought stress is miR156. MiR156 has been upregulated under drought stress (Santhi et al. 2016). In Arabidopsis studies, it has been found that miR156 targets SPL genes (Wang 2014). Squamosa-promoter binding protein-like (SPL) forms several transcription factors essential for various plant-related growth and development (Yu et al. 2020). There are not enough reports of functional characterization of SPL family genes. However, in Medicago sativa, it is found that SPL9 gene overexpression can induce drought tolerance in plants (Hanly et al. 2020). Another gene of the SPL family, SPL13 is also found to regulate stress-responsive mechanisms (Arshad et al. 2017). It regulates root architecture to be thicker, denser, and longer in size to reduce water loss under drought stress (Hanly et al. 2020). Some studies also reported the antioxidant activity under drought stress also is regulated by the SPL family (Arshad et al. 2017). The catalase activity under mild drought stress is enhanced to combat ROS production, which leads to the involvement of miR156 (Hanly et al. 2020). Another study

reports that SPL9 increases anthocyanin production under drought stress (Cui et al. 2014). Anthocyanin accumulation under drought stress works in two ways: first, they provide red color to the leaf hence protecting the plant from excessive sunlight, and second, regulate stomatal transpiration to reduce water loss (Cirillo et al. 2021). Therefore, based on all reports in the past, I have hypothesized a pathway of SPL that can be responsible for the tolerance of turmeric under drought stress, which is miR156-SPL pathway (Figure 29).

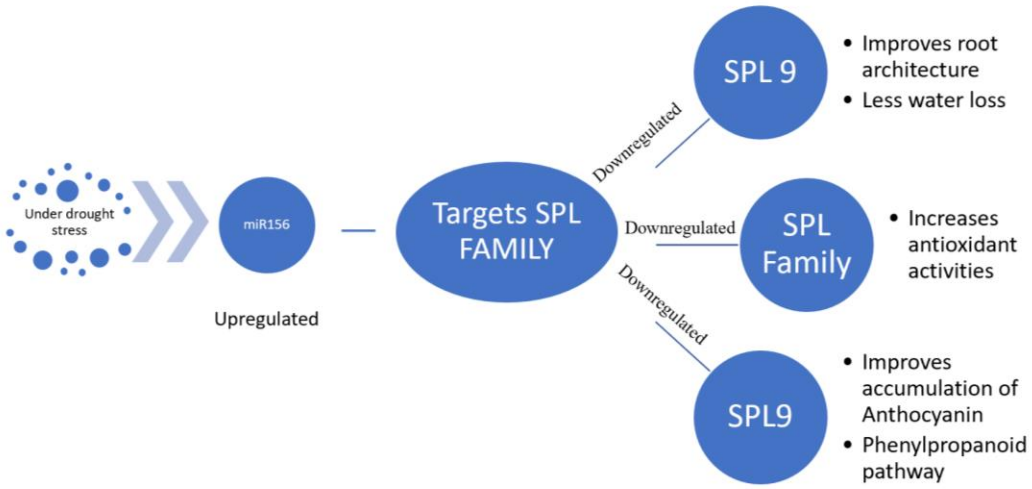


Figure 29: Hypothesized pathway of miR156-SPL pathway under drought stress.

Another miRNA that has been upregulated under drought stress in turmeric plants was miR167 (Santhi et al. 2016). And this miRNA targets Auxin Response Factors (ARFs). Different proteins from ARFs regulate soluble sugar content, root development, and chlorophyll content under drought for plants to adapt (Verma et al. 2022). Usually, in dicot plants, the downregulation of miR167 enhances the accumulation of the auxin response factors and thus improves shoot and leaf development under drought stress

(Bouzroud et al. 2018). So, I hypothesized that the auxin regulation pathway also gets affected under drought stress in turmeric plants (Figure 30).

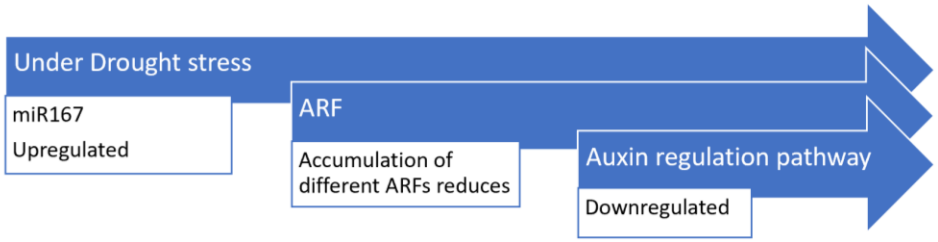


Figure 30: Hypothesized pathway of miR167-Auxin Regulatory pathway under drought stress in Turmeric.

A recent study found 188 Multiple Signs of Adaptive Evolution (MSA) in *Curcuma longa* genome. 177 of them are related to secondary metabolism pathway or defense responses to biotic and abiotic responses (Chakraborty et al. 2021) (Figure 31). Six genes from plant U-box E3 ubiquitin ligase (PUB) protein are found in the MSA gene collection of turmeric (Chakraborty et al. 2021). These PUB proteins, only a few, are functionally analyzed, and their roles range from self-incompatibility to different stress responses to biotic stress and abiotic stress, such as drought stress (Yee et al. 2009). Another gene, chalcone isomerase (CHI), is found in MSA gene collection of turmeric and plays a vital role in the anthocyanin biosynthesis pathway (Chakraborty et al. 2021). The chalcone synthase family (CHS) is responsible for the first committed step of the phenylpropanoid pathway (Wu et al. 2020). Now, it has been found that this pathway downregulates under drought stress due to overexpression of miR156 in turmeric (Santhi et al. 2016), which reduces another secondary metabolite production, flavonoids (Wu et al. 2020). Four genes are responsible for converting coumaroyl-CoA from phenylalanine, leading to two biosynthesis pathways: the Curcuminoid biosynthesis pathway (Rodrigues

et al. 2015) and the phenylpropanoid biosynthesis pathway (Zhu et al. 2015). Further, the synthesis of curcumin from coumaroyl -CoA takes place in the presence of 4 enzymes: *CURS1* (curcumin synthase 1), *CURS2* (curcumin synthase 2), *CURS3* (curcumin synthase 3), and *DCS* (diketide-CoA synthase), which are again from CHS family. With all past reports, I have hypothesized another pathway that gets affected (Figure 32).

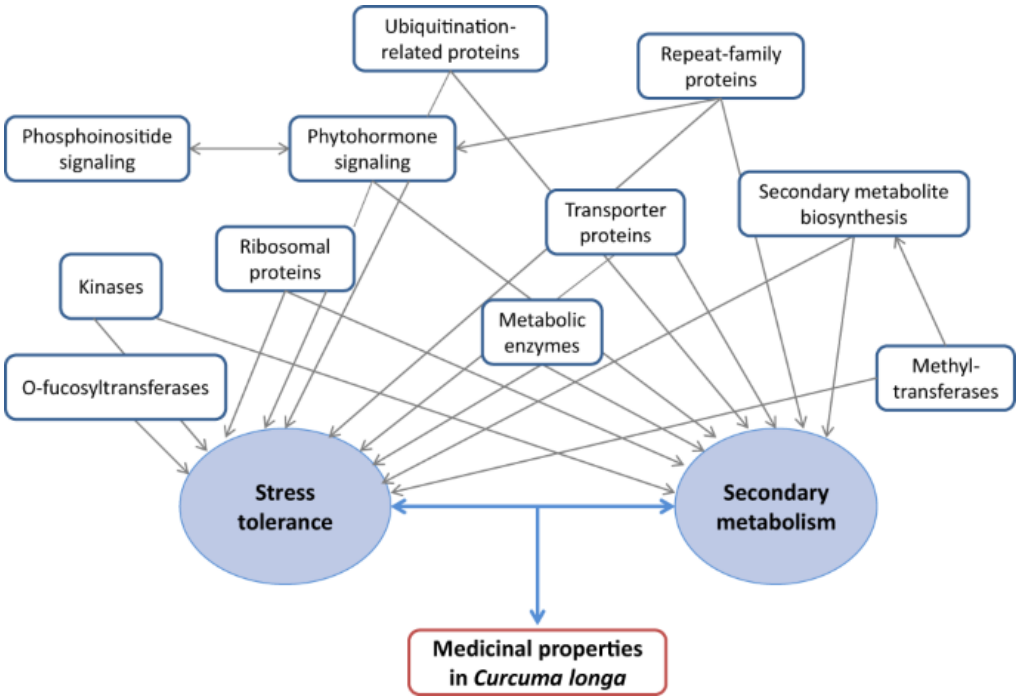


Figure 31: Functional characterization of MSA genes in the genome of turmeric (Source: Chakraborty et al. 2021)

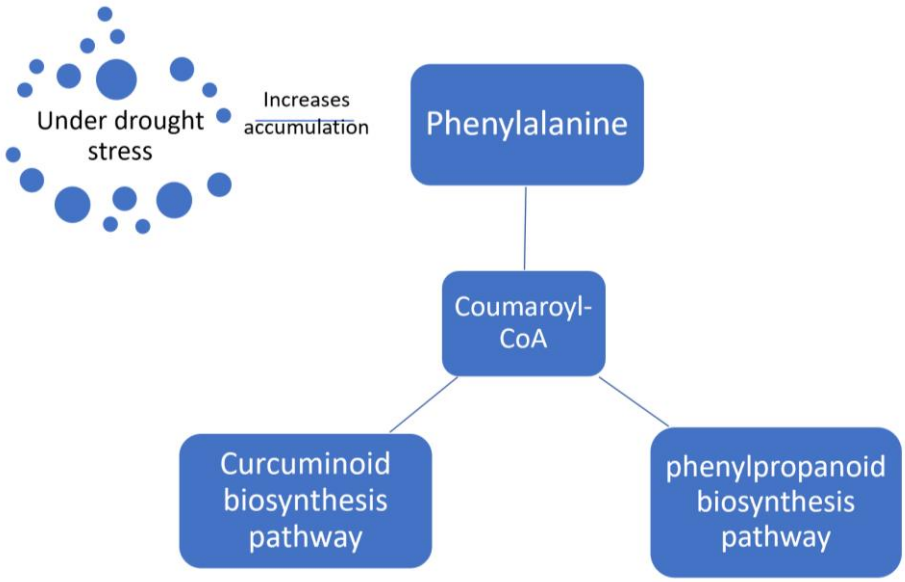


Figure 32: Hypothesized pathway affected by drought stress in turmeric

Based on past and current studies, it is established that turmeric has excellent potential for pharmaceutical and culinary use. With time, the demand for turmeric in Western countries has increased for different purposes. Understanding the changes in physiological & molecular levels will help to know the potential of drought tolerance in C*urcuma longa.* This study can lead to a genetically improved turmeric plant with high drought tolerance with better production.

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